

Electronics in

74. Electronics in Medicine

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Electronics has long made an undeniable and valuable contribution in the field of medicine. New medical electronic devices are based on available medical knowledge combined with technologies available in the electronics field. Electronics in medicine has a wide range of applications, from diagnostics to therapy, always aiming to provide new tools to improve the well-being of the population.

Nowadays, miniaturization and integration technologies are driving the development of new electronic medical devices that should be portable, wearable, and unobtrusive, but with controlled costs. Regardless of the technologies involved, usually the main goal of such devices is the measurement of physiological variables for use in diagnostics, monitoring, and therapy.

Therefore, this chapter starts with the description of several signals that are commonly recorded, the corresponding electronics and sensing mechanisms used, and problems associated with such acquisition systems, focusing on wireless and wearable devices. It ends with examples of electronic devices in development essentially for remote health monitoring.

This chapter is not about electronics nor on their design, since there are many good books on this topic already, written either specifically for the (bio)medical field or for the general field of electronics. The aim of this chapter is rather to present to the reader those aspects that are common and specific to many medical electronic devices, and to give an overview of what is required and the challenges faced when designing electronics in this field as well as a few such examples under development.

74.1	Basics	1338
74.1.1	Fields of Application	1338
74.1.2	Designing Electronics in Medicine	1339
74.1.3	Medical Electronics Specific Requirements	1339
74.2	Electronic Sensing	1341
74.2.1	Health Condition Monitoring	1341
74.2.2	Biosignals and Transducers	1342
74.3	Electronics for Wireless Health Monitoring	1344
74.3.1	Requirements for Wireless Devices	1345
74.3.2	Data Acquisition	1346
74.3.3	System Integration	1347
74.3.4	Wireless Communications	1348
74.4	Power Supply	1348
74.4.1	Battery Power Budget Considerations	1349
74.4.2	Wireless Power	1350
74.4.3	Wireless Power Link Analysis	1350
74.4.4	Wireless Power Charger	1351
74.4.5	Energy Harvesting	1352
74.5	Wearable Medical Electronics	1353
74.5.1	Wearable Systems	1355
74.5.2	Categories of Wearable Systems	1355
74.5.3	Design Requirements	1356
74.5.4	Sensors for Wearable Systems	1357
74.5.5	Sensing Methodologies	1357
74.6	Electronics in Medicine at Work	1358
74.6.1	Wireless and Wearable Low-Power Health Monitoring Systems	1359
74.6.2	Sensors	1359
74.6.3	Wireless Link	1360
74.6.4	Optical Biopotential Recording	1360
74.6.5	Optical Electrodes	1360
74.6.6	Optical Signal Acquisition	1361
74.6.7	Localization Solutions	1361
74.6.8	Ambient Assisted Living Applications	1363
74.6.9	Wireless Link Design for Biomedical Applications	1366
	References	1373

The definition of electronics in medicine has many facets and is hard to define. We may therefore start with the question: What is *electronics in medicine*? Is electronics not always electronics? We use the following definition [74.1, 2]:

Medical electronics – A branch of electronics in which electronic instruments and equipment are used for such medical applications as diagnosis, therapy, research, anesthesia control, cardiac control, and surgery.

74.1 Basics

From our point of view, electronics in medicine encompasses all aspects of electronics that deal with the problems found when electronics are used to understand or solve a medical problem, not necessarily restricted to humans. Electronics in medicine may include the study of the electrical behavior of the human body and how it responds to electrical stimuli. Those stimuli may be in the form of voltages, electric currents, electric fields or magnetic fields. In this case, electronics is used as a model to represent such behavior.

Electronics may be used as a solution to describe biological activities, as happens with the widely used Hodgkin–Huxley model for a neuron cell [74.3], as a solution to help restore lost electrical activity, as is the case with pacemakers [74.4–6] or Parkinson disease [74.7, 8], to help in the replacement of full human body parts [74.9–11], for diagnostics [74.12–15], for therapy [74.16], or for monitoring [74.17–19].

In this sense, it is difficult to write a chapter that includes all the aspects, difficulties, and aspects associated with electronics in medicine. Moreover, there are different perspectives and different views when looking at medical electronics, corresponding to electrical engineering, chemical engineering, ethical aspects, security and privacy issues, physics, and medicine, to mention just a few. However, it is possible to identify two different sides of the same coin: electronics and medicine. On the electronics side, no one is very interested in knowing what is going on behind the electronic system under development. Rather, the system is purely developed to fulfill a set of requirements. From the medical side, what is important is how the system can help to solve an existing problem and how the system can be used, e.g., *please give me a system that enervates the bladder, when placed inside the spinal cord, close to C4–C5*. Fortunately, these two worlds are getting closer, and many great things are expected from electronics in medicine.

Since this field includes many applications, from magnetic resonance imaging (MRI) to electrocardio-

gram (ECG) recording, through biosensors and DNA chips, this is a very broad field, and it is impossible to include and describe all the electronics used in medical applications. Therefore, this chapter introduces the reader to several issues that must be addressed when developing electronic devices for application in medicine. There are many problems that must be tackled and solved when using electronics in medicine. Of course, some of these problems, fortunately, are also shared by other application fields.

74.1.1 Fields of Application

The most obvious application of medical electronics is their use for medical or clinical applications (diagnostics, treatment, and monitoring). However, electronics developed for these applications may have other important fields of applications. One application that is closely related to medical applications is human augmentation, which may apply at different levels, of which two can be highlighted: sensorial [74.20–24] and physical capabilities augmentation [74.23, 25, 26]. For the former, we can give the example of brain–computer interface (BCI) systems that can be used to detect what is happening in the field of vision of a soldier, before he is able to become aware of it [74.27–30]. For the latter, we can give the example of exoskeletons, which can be used to help soldiers in transportation of heavy loads (as in the movie *Avatar*) during long walking periods, or to help an old person with problems in locomotion.

Another very important field of electronics in medicine is its use for ambient assisted living (AAL) applications (Chap. 67, [74.31]). Electronics are used to record, understand, and control those environments. The main idea is to create smart environments that can sense several physical variables to allow implementation of control and decision loops, making that environment *smart*. Such smart environments may be used, e.g., for environmental control, or, focusing on the medical field, to help people with their daily activities.

One major problem they try to tackle is to deal with the ageing of the world population [74.32], for which new innovative technological solutions are expected from the field of medical electronics.

Electronics in medicine may be also used for performance improvement [74.33, 34] and performance monitoring [74.34, 35, 35–37]. These last two applications are closely related, since performance monitoring is a way to understand what can be done to improve it. Monitoring of performance can be done by measuring the physiological signals of the user, or by measuring the outcome of the human action.

Last but not least, many advanced electronics developed for application in medicine may be used for entertainment and interaction purposes [74.23, 38, 39]. All devices developed to monitor posture, position, or muscle signals, or brain activity, can be used as new paths to control new interfaces, leading to a new concept of human–machine interfaces [74.21, 23, 33].

74.1.2 Designing Electronics in Medicine

The design of electronics for medical applications implies the fulfillment of a set of requirements due to the nature of their operation. Such electronics must be designed to be small and capable of being integrated into the environment where they will operate. Another important aspect is that these systems may be required to operate in very harsh environments (inside the stomach, where lots of substances are prepared to digest almost everything), using only a single battery, with no possibility of replacement during tests, and many times only possible using surgical procedures. This requires the use of low-power design techniques, requiring biocompatibility and long-term usage capability.

Since in many applications, the electronics are required to interface with the human body, in order to record signals or to deliver signals or drugs, or to monitor their effect on the human body, one feature of such electronic devices is the ability to register very small variations of the variables under measurement, and many times it is necessary to record many signals at the same time, leading to challenges in terms of analog-to-digital conversion. Most electronics that are used in medicine are designed to sense and transmit a variable, or to control an actuator. In this way, most of the developed solutions have their roots in instrumentation or communications electronics.

Electronic devices in the medical field are usually based on state-of-the-art technologies in order to maximize benefits. They make use of the latest developments

in terms of micro/nanoelectronics, micro/nanosensors, smart material technologies, device integration, device packaging, smart systems, etc. However, beyond all the electronics, most intelligent devices will only succeed if they are also designed to comply with the requirements of the software agents that are used to monitor, detect, and provide alerts based on the measured health condition.

Another important requirement is the ability to operate as a clinical tool. Many electronic devices being developed that claim potential use in the medical field are far from this goal, because there is a big difference between developing a device to monitor the heart rate while jogging versus use in a clinical application. Before clinical use, a long process of certification has to be completed [74.2, 4, 40]. Simultaneously, many such electronics must also be reliable for operation inside a clinical environment, such as a hospital. As is well known, mobile phones are certified electronics devices, but they are not allowed inside many clinical facilities. This is also the situation for many wireless technologies.

Beyond the certification requirements, which are more demanding than in other fields of electronics, the development of electronics for medicine poses its own technical challenges, due to the various operating constraints. Also, legal and cultural aspects, that differ from country to country, have to be dealt with. Moreover, since the developed solutions are, usually, for use by medical doctors, it is also necessary to break the acceptance barrier for some new technologies, despite the certification. Finally, and not less important for many applications, the proposed electronic solutions must gain user acceptance, in particular if they are for use outside the clinical environment.

74.1.3 Medical Electronics Specific Requirements

At first glance, the field of medical electronics may not seem as challenging as it really is. Signal amplitudes and frequency ranges are not that demanding, the operating range is not that large, the operating voltages and currents are, in general, normal, etc. This leads to the use of standard amplifiers, filters, analog-to-digital converters (ADCs), digital-to-analog converters (DACs), microprocessors, microcontrollers, etc. So, what are the problems when designing medical electronics? The problem lies, as in all other electronic fields, in the combination of the requirements specific to such devices.

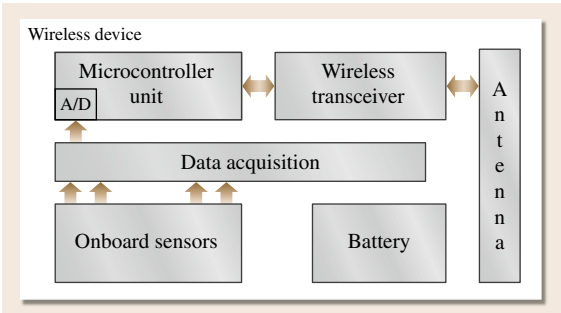


Fig. 74.1 Main electronic components required for advanced clinical data acquisition

Figure 74.1 shows a schematic of the electronics involved in a typical solution for a biomedical device. This represents a solution for an advanced device, meaning that only the most sophisticated devices require all the modules represented. In other cases, only a subset of such modules will be required; for example, not all electronic modules used in medicine require use of a microcontroller or a bidirectional wireless link.

Bearing in mind the discussion above, a set of challenges may be identified when developing electronics for medicine. These challenges may be found in the areas of data acquisition, filtering, power autonomy, and communications. Furthermore, for AAL applications, it is desirable that devices may be easily worn by the user, leading to miniaturization and wearability requirements. This has also implications in the fashion field, because many people do not like to use electronic devices because they are not fashionable.

The challenge at the data acquisition level results from the combination of a set of requirements particular to this type of systems. The frequency range of interest includes the 50/60 Hz frequency used by power lines. This is a significant problem at various levels. When designing power supplies for operation with these devices, special care is required since the direct-current (DC) voltages are obtained from a 50/60 Hz alternating-current (AC) voltage that will introduce a very strong oscillating component into the supplied power. This interfering component may easily become larger than the signal to be measured, and the minimum signal amplitude that can be measured will be increased. To avoid this, either very careful design of the power supply is required, or a battery is used as the power supply. Since many systems are designed for use on-body and in-body, the battery solution solves many of these problems. However, power lines may also be coupled to the human body, and when measuring biopotentials, inter-

ference is still a problem even when using batteries as the power supply. It is very easy to have a few millivolts superimposed on the measured biopotential due to the power lines, which is of the same order of magnitude as the biopotential itself. One possible solution is to use a notch filter to remove this component. However, this is a challenge because such filters must have a very high quality factor in order to remove only the 50/60 Hz component, because the signal components of interest are around this frequency, and they must have also a high attenuation factor. Attenuation of the order of 30–40 dB is sufficient for most applications, although some may benefit from values up to 70–80 dB. However, this high attenuation introduces a new challenge, namely on how to obtain such devices with very low noise factor. Most designs with off-the-shelf components only enable devices with noise on the order of a millivolt.

Another problem during signal acquisition is associated with subject activity, which has frequency components inside the frequency band of interest, introducing so-called movement artifacts. Since these slow components are inside the band of interest, they cannot simply be removed by standard low- or high-pass filtering. More complex filters, such as adaptive filtering, must be used to remove these components.

At the acquisition level, we have another important aspect, especially for biopotentials, which is the well-known amplifier common mode rejection ratio (CMRR). Since the subject under measurement is at a different potential with respect to the power supply, often the output signal may be a few millivolts, even when both inputs are at the same potential (common potential). The standard solution is to connect the subject to a reference potential, thereby lowering the common mode voltage and reducing its effect on the recorded signal. However, this solution requires the use of an extra electrode, requiring also extra room to accommodate it, which is not desirable when designing miniaturized devices.

Power autonomy is also a very important issue and is easy to understand when we think about other power-limited devices, such as mobile phones. When dealing with medical electronics, this issue may become more complicated, since some devices are required to operate inside the human body for a long period, as happens with pacemakers. In terms of power autonomy, we may divide the systems into three categories: systems to go inside the body (such as endoscopic capsules), systems to operate inside the human body (such as pacemakers), and systems to operate on (outside) the body (such as

wearable devices for AAL). Usually, when recording a single channel [electroencephalogram (EEG), ECG, etc.], it is not difficult to find a solution that operates on a small battery for a few days. The problem is when it is required to record many channels (such as for EEG in epilepsy applications), or when it is also required to actuate another device to obtain the required reading (e.g., a sphygmomanometer). When readings from many channels are required, one option is to design all the electronics based on low-power approaches. The other option, which is also valid for situations where actuators are also present, is to integrate energy harvesting/scavenging technologies. Despite some promising proposals, such approaches are not yet available as solutions to fully power a complete system. However, they may be a potential solution to extend the autonomy of

existing systems, and in the near future it is expected that they may replace batteries in some applications.

We have also to consider communications, in which regard the desirable solution is usually to use wireless communications. However, this introduces new challenges, because this subsystem is typically the most power hungry. To reduce the power consumption of this subsystem, an as efficient as possible power amplifier design must be used, and antennas should also be selected as efficient as possible [74.41,42]. The challenge regarding antenna design results from the operation of wireless systems in low frequency bands in order to decrease the power required for transmission, but which degrades antenna efficiency because they must be electrically small to fit inside the space available in the device.

74.2 Electronic Sensing

Electronics in medicine may include many applications, as mentioned above; for example, ultrasound and MRI imaging systems are often used in diagnostics, as are ECG recording systems, all of which look very different at a high level. However, looking in more detail, all of them are designed to sense physical variables to aid clinical diagnosis. In this way, in most medical electronic devices, the requirement is to design adequate instrumentation electronics to interface with a transducer that is used to convert a physical variable of interest into an electrical signal, which is finally used to help in the diagnostics.

The success of electronics in medicine relies on their ability to help medical doctors in the determination of a set of parameters that are compared against a range of values considered as healthy. After beating the challenge of determining which parameter is important to measure, and in which range it must be, it is up to the medical electronics to provide solutions to perform these measurements reliably and in a solution that can be easily deployed. In this way, instrumentation is a very important field when dealing with medical electronics. Many medical devices deal with the measurement of specific variables, which may be chemical, physical or electrical. These measurements are supported by appropriate electronics, as well by a transducing mechanism to convert, e.g., an ionic concentration to a voltage, proportional to it.

Some recordings may be obtained by direct measurements, as for measuring body temperature, where

a temperature sensor is used. Alternatively, they may be obtained by indirect measurements, such as measuring breathing rate using a temperature sensor, where temperature changes of air flow are measured to determine the breathing rate. In this way, sensing techniques and sensors are also very important fields when dealing with medical electronics. They are essential, because without sensors there is no medical information to process, analyze, transmit or store.

74.2.1 Health Condition Monitoring

Health condition monitoring requires the use of a set of sensors to monitor a wide variety of signals, either clinical or environmental, forming a sensing component, for example, surface electrodes to measure bioelectric signals, together with the acquisition electronics.

The measured signals can be classified according to their nature and/or type of monitoring. Figure 74.2 presents a few different categories of signals that are commonly measured by sensors, as well as the main parameters assessed.

Physiological signals include bioelectric events and other biochemical and physical parameters that are crucial for assessment of the user's state of health. They can be detected by means of biosensors, which are a particular category of wearable sensors that have the ability to monitor biological properties and processes.

Each signal is important for capturing the entire status of the subject and the surrounding environment. On

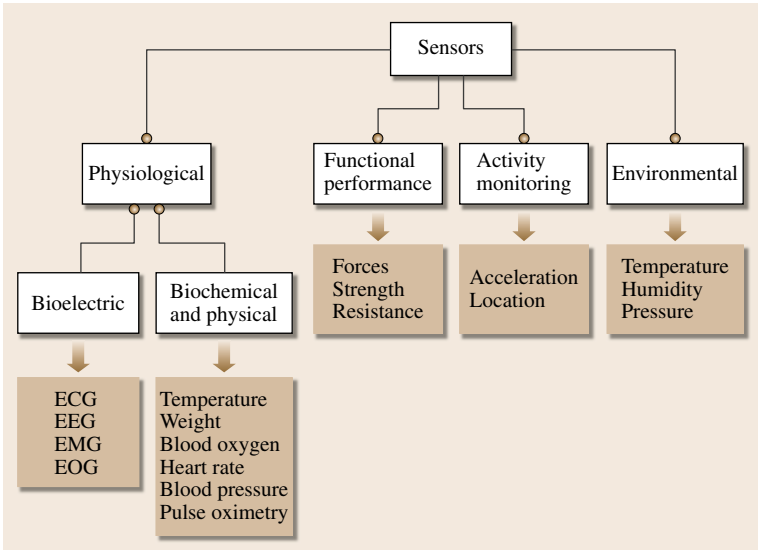


Fig. 74.2 Categories and types of possible biosignals measured to infer an individual health condition

top of such acquisition systems, use of *intelligence* is usually required so that the system can automatically detect any relevant health condition and, based on that, trigger alarms or send warnings to healthcare providers or the patient being monitored [74.19, 43].

74.2.2 Biosignals and Transducers

Biosignals offer an interface between the human body and electronic systems, and they are directly related to the physiology and function of organs and organ systems, hence allowing evaluation of the health condition of a subject through their acquisition and monitoring.

An electronic system is used to measure physiological signals. Those signals may be electrical, or if not, they are translated to electrical signals using one of various transducer or translation methods.

Table 74.1 presents examples of the most commonly monitored biosignals for the human body, and their nominal values [74.44–46].

Table 74.1 allows a few conclusions to be drawn regarding the signal and transducer characteristics involved in the process of health monitoring. A few that are important to highlight are: signal amplitude, frequency, power, and the signal translation mechanism.

Signal Amplitude

Bioelectric signals are not difficult to measure from this point of view. From Table 74.1 we can see that the most demanding signals are in the μV range, which, when compared with the radiofrequency (RF) signal at an an-

tenna input (in the range of pV), is very high. When looking for nonelectrical signals, it may also be observed that both signal amplitudes and their dynamic ranges are not very challenging.

One main problem related with signal amplitudes lies in the fact that different signals are in the same range of amplitudes and frequencies. This is a problem since some signals tends to act as interference to others, thus appearing as so-called artifacts. As an example, electrocardiography (ECG) or electrooculography (EOG) signals usually appear overlapped on electroencephalography (EEG) signals. Even simple patient movement, which occurs on the order of a few Hertz, interferes with signals such as ECG, respiration rate, and blood pressure.

Signal Frequency

From this point of view, physiological signals are also not difficult to measure. The maximum frequency is on the order of a few kilohertz. However, since many of them are in the same frequency band, as mentioned above, they tend to interfere with one another. Another common problem is electromagnetic fields coming from power lines (operating in the frequency range 50–60 Hz), which are easily coupled through the power source (that may also introduce a 100 Hz component due to the use of full-wave rectifiers), or because the human body acts as an antenna for those fields. Sometimes the coupled signal may have higher amplitude than the biosignal being measured. One solution is to use batteries to power some electronic devices, avoid-

Table 74.1 Description and properties of commonly monitored biosignals

Biosignal	Range	Frequency	Sensor type
Blood flow	1–300 ml/s	0–20 Hz	Electromagnetic or ultrasonic
Blood pressure	0–400 mmHg	0–50 Hz	Cuff or strain gage
Blood oxygen saturation	95–99%	0–50 Hz	Paired infrared LED and photodiode
Cardiac output	4–25 l/min	0–20 Hz	Fick, dye dilution
Electrocardiogram (ECG)	0.5–4 mV	0.05–150 Hz	Skin electrodes
Electroencephalogram (EEG)	5–300 μ V	0.5–150 Hz	Scalp electrodes
Electrooculogram (EOG)	10–100 μ V	0–10 Hz	Skin electrodes
Electromyogram (EMG)	0.1–5 mV	0–10 kHz	Needle electrodes
Electroretinogram (ERT)	0–900 μ V	0–50 Hz	Contact lens electrodes
Action potentials	–80–80 mV	10–10 kHz	Neuronal electrodes
Local field potentials	0.1–1 mV	0.1–500 Hz	Neuronal electrodes
pH	3–13 pH units	0–1 Hz	pH electrode
$p\text{CO}_2$	40–100 mmHg	0–2 Hz	$p\text{CO}_2$ electrode
$p\text{O}_2$	30–100 mmHg	0–2 Hz	$p\text{O}_2$ electrode
Pneumotachography	0–600 l/min	0–40 Hz	Pneumotachometer
Respiration rate	2–50	0.1–10 Hz	Impedance, electroactive
Temperature	32–40 °C	0–0.1 Hz	Thermistor
Activity		0–100 Hz	Accelerometer, GPS, Wi-Fi
Body weight	0–150 kg	1 time/day	Weighing machine
Glucose	70–180 mg/dL 3–8 mmol/L	1–5 times/day	Glucose meter

ing interference from the power supply. To remove the effect of interference picked up by electromagnetic coupling, the solution is to use an instrument amplifier not with an input impedance as high as possible but rather designed to *kill* the 50/60 Hz component. However, the bioelectric signal has limited power. Another problem, due to the low frequencies involved, is the requirement for high capacity values, leading to large capacitors for filtering blocks. This has implications in terms of device dimensions and response time for the transitions of such systems.

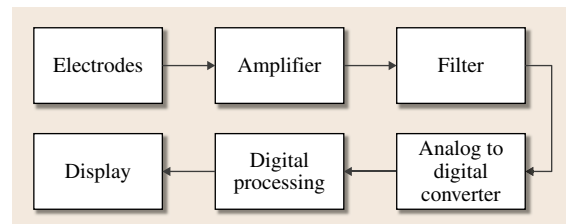
Signal Power

Many of the important signals that need to be measured are biosignals that originate from body electrical activity. One characteristic of such signals is their small amplitude. However, that is not the main problem with their acquisition; the main problem is their low power. Figure 74.3 shows a typical acquisition system used for bioelectric signals. Figure 74.3 shows the bioelectric signal source, the electrodes (which are responsible for the interface between the ionic human body system and the electronic system), and a high-input-impedance amplifier, usually an instrumentation amplifier. Figure 74.4 shows the equivalent electric circuit.

Since the source current is on the order of pA, the available power is also very small. This is a problem when implementing the proposed solution for power line noise cancelation. The interference is canceled, but the biosignal is also attenuated. This means that any small current flow to the instrumentation amplifier will lead to a voltage drop on the electrodes, which will make the available voltage even smaller and more difficult to measure.

Signal Translation

When the biosignal to be measured is not an electrical signal, it is necessary to use a transducer to obtain an electric signal proportional to the physical variable being measured. Oximetry provides a signal obtained

**Fig. 74.3** Typical acquisition system for bioelectric signals

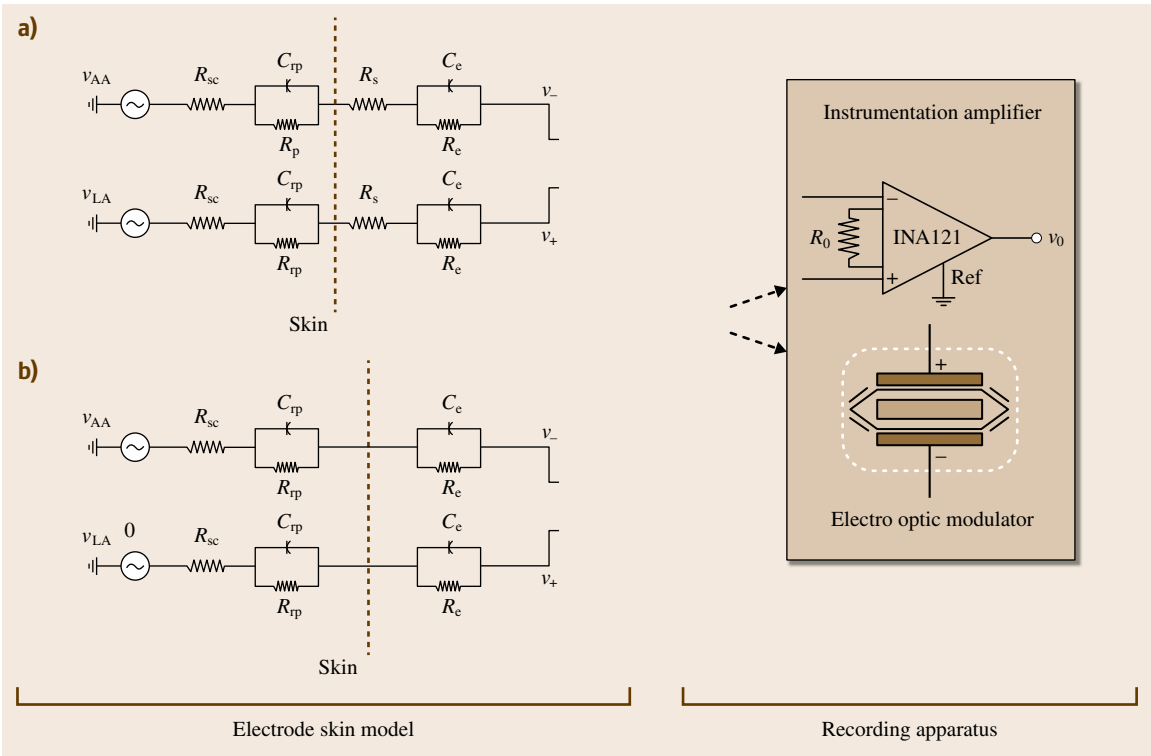


Fig. 74.4a,b Equivalent circuit for biopotential transduction using (a) wet/dry electrodes and (b) capacitive coupled electrodes

from measurements of light, spirometry results in a signal obtained from air flow, and the respiration rate may be obtained from temperature variation or by abdominal volume variation. These are only a few examples, but one key aspect may be highlighted: most of these devices require also an actuator and they tend to be bulky.

In this way, measurements become more difficult when portable devices are required, and even more difficult when wearable devices are considered. When portable or wearable devices are required, often it is difficult to find a solution that is reliable enough and at the same time acceptable for use by the patient.

74.3 Electronics for Wireless Health Monitoring

Wireless health monitoring systems (WHMS) are among the most promising applications of wireless communication technology, and they have been a very active research topic with considerable advances during the last decade. These systems consist of groups of wireless-enabled sensors attached to the human body in a noninvasive or minimally invasive fashion, together forming a wireless body-area network to acquire diverse physiological signals (also known as biosignals) and transmit them via radio to a remote location, for display or monitoring to enable assessment of the user's health

condition. Thorough reviews and surveys on WHMS can be found in [74.47–49].

The development of WHMS envisions the improvement of the quality and efficiency of healthcare, which is currently facing increasing demand and, consequently, increasing global costs. The tendency for increasing healthcare demand is mainly motivated by the ageing of the world population, which is expected to cause an increase in the incidence of chronic diseases and physical disabilities [74.47]. According to the United Nations, the world population is currently

enduring an unprecedented and irreversible ageing process, which will cause a growth in the proportion of elderly people (over 60 years old) to a third of the population in developed countries, and to 22% of the total world population by the year 2050 [74.50]. Therefore, there is a growing need for health monitoring solutions to provide more efficient use of healthcare resources to caregivers and better quality of life to the elderly or diseased, relieving excessive pressure on healthcare demand and costs.

Use of wireless technology in healthcare provides new means for close observation of the health condition of patients. Typically, observation of a patient takes place when he feels some sign of illness and reactively seeks healthcare. Alternatively, his health condition may be followed by a caregiver, requiring the patient either to visit a healthcare facility or to stay there in case of prolonged examination or intervention. This leads to time and location constraints for the patient and can only provide limited periods of observation of his health condition, which may mean that diagnosis of health complications or pathologies may be late or overlooked. Moreover, conventional health monitoring equipment uses wired connections to the patient, hindering the activity of monitored patients.

On the other hand, WHMS inherently provide mobility to the patients, allowing them to be monitored, virtually, anytime and anywhere within the range of the wireless network. Hence, WHMS gives patients the possibility to be monitored in a convenient location, typically at home, with reduced disturbance to their daily activities, helping them to adopt a healthier lifestyle and improving their quality of life and independence. In addition, WHMS also offer the possibility of continuous health monitoring. Following the current trend towards miniaturized and low-power-consumption electronics development, WHMS can be used for long periods of time (e.g., 24 h), providing approximately continuous evaluation of the health condition of a subject. Furthermore, continuous health monitoring is a very useful feature for enhancement of preventive care and quality of treatment. By monitoring a subject continuously, any slight changes in health condition in relation to the long-term circadian pattern can be detected at an early stage by the caregiver, allowing for better decision-making and anticipating the occurrence of severe health complications. Thus, continuous health monitoring offers better treatment possibilities for patients suffering from chronic diseases or enduring rehabilitation processes. Subjects without illness may also be continuously monitored to receive feedback on their activities and modify

their behavior to prevent illness by the promotion of a healthier lifestyle.

74.3.1 Requirements for Wireless Devices

Even though WHMS have the potential to provide great benefit in healthcare, they have not yet matured to a point of general use, as their application to real-life situations and acceptance by caregivers and patients is hindered by a considerable number of application and user requirement issues, including the following:

Ergonomics and Wearability

A WHMS needs to be as small and lightweight as possible so that the user feels comfortable when using it over long periods of time, and it should not affect the patient's daily activities. Ideally, the WHMS should go unnoticed by both the user and others, as the users of WHMS may become stigmatized if it is perceivable. An appropriate solution is integration of the WHMS into textile garments, resulting in wearable or smart-textiles monitoring systems [74.49, 51].

Low Power Consumption

WHMS, as a mobile application, are required to include their own power source, which is typically in the form of a battery, although energy harvesting solutions appear as a possible alternative in the future [74.52]. The power sources of these devices can only offer limited operation time, and therefore it is essential to reduce the power consumption to minimal levels, to ensure appropriate monitoring periods.

Quality of Service (QoS)

The wireless communication protocol implemented on the WHMS must include a proof mechanism to ensure that the biosignal data are reliably delivered to the base station, especially for continuous monitoring and if an emergency situation is detected.

User Friendliness

WHMS operation should be autonomous but perceivable by the user of the system, either the patient or healthcare personnel, and should require as little maintenance as possible, in order to achieve user acceptance.

Privacy and Security

The health condition of a patient must be regarded as private information, therefore data encryption methods and protection against unauthorized access should be implemented.

WHMS are expected to be a valuable tool to address the effects of population ageing on healthcare. They provide benefit to both patients and caregivers, enabling improved quality of life and independent living for patients and improved efficiency and better use of healthcare resources to caregivers. However, the design of such systems is a challenging task, and solutions able to reach wide acceptance by patients and caregivers are yet to be implemented.

74.3.2 Data Acquisition

Acquisition of biosignals from the human body requires similar instrumentation and methods as for any other signal which requires transducers. However, biosignals, namely those which are electrical, typically have small amplitudes and bandwidths and hence are very susceptible to interference or artifacts, which can easily be superimposed on the biosignal to be monitored. Moreover, part of the information contained in the biosignal may not be readily available from the raw signal, requiring further analog or digital processing for feature extraction [74.53]. Therefore, the design of acquisition electronics for biosignals must contain suitable instrumentation to ensure the integrity of the information contained in the biosignal, while any forms of interference or artifacts must be minimized, so that the acquired biosignal is useful for health condition monitoring or diagnostics, since a corrupted signal could provide misleading or meaningless information. Figure 74.5 shows a generic block diagram of instrumentation for biosignal acquisition.

Sensors, represented in the first block, are responsible for the conversion of clinically relevant physical quantities into an electrical output that can be processed by electronic instrumentation. Different sensor types are used according to the physical quantity associated with the biosignal. For electrical biosignals, it might be presumed that sensors would not be required since an electrical property, namely current flow on the human

body, is measured. However, transducers are indeed required for acquisition of electrical biosignals, since the current flow on the human body is due to ion transport whereas current flow in the acquisition instrumentation is due to electron movement. This conversion is provided by biopotential electrodes [74.53], which are placed on the surface of the human body, as close as possible to the organ that generates the biosignal, such as the heart (ECG) or brain (EEG). For nonelectrical biosignals, various types of sensors are used depending on the nature of the biosignal, such as mechanical (e.g., sphygmomanometer for blood pressure monitoring) or thermal (e.g., thermocouple for temperature measurement) devices, and the locations of sensor placement vary according to where the signal is strongest or the measurement most accurate.

Following the sensor stage, the signals obtained are usually weak in amplitude and may have considerable levels of added interference, as mentioned above. Thus, analog processing, featuring low-noise amplification and filtering, is required to increase the amplitude of the acquired signal, match the input range of the analog-to-digital converter, and limit its bandwidth to the range of interest of the monitored biosignal. Care must be taken when designing or choosing amplifiers for analog processing of biosignals. Desirable features for a biosignal amplifier include [74.45, 54] high input impedance (about 10 MΩ to minimize the current drawn from the monitored signal and hence signal distortion), low output impedance (to provide as much power as possible to the load), high differential gain (10³ or greater), high common mode rejection ratio (to minimize electromagnetic interference and electrical noise), and frequency response temperature stability. Filtering of the biosignal is usually required, since biosignals are susceptible to interference and artifacts. Filtering of biosignals can be performed in either analog or digital fashion. Analog filters are implemented using modified amplifier stages that attenuate unwanted frequency components from the acquired biosignal while provid-

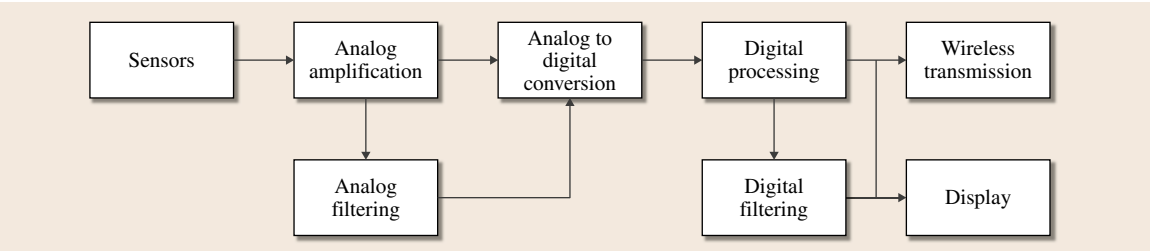


Fig. 74.5 Typical block diagram of the necessary instrumentation for biosignal acquisition

ing gain to those which are relevant, thus contributing to removal of low- and high-frequency noise. If necessary, electromagnetic interference from power lines may be further attenuated using notch filters centered at 50 or 60 Hz. Besides, analog filtering is also used to limit the bandwidth of the monitored signal according to the sampling frequency of the analog-to-digital converter, thus avoiding aliasing. Although analog filters are essentially used to remove undesirable signal features, their design must ensure that the filter itself does not cause distortion of the relevant frequency components of the monitored signal. To minimize signal distortion, analog filters should be designed to provide a constant modulus and linear phase frequency response over the pass band [74.45, 53].

After the analog filtering stage, the monitored signal is converted to digital format using standard sampling and quantization methods such as sigma-delta or successive approximations [74.55]. In digital format, biosignals can be handled by digital processors, such as microcontrollers or microprocessors, and be further processed using digital filtering techniques. Digital filters offer advantages over analog filters, as they can be easily modified by changes in programming code, as opposed to changing components in analog designs, and they do not suffer from component tolerances or environmental influences such as temperature. Still, digital filtering and processing techniques require that the processor has enough available computational power, depending on the complexity of the algorithms to be executed by the processor. This may not be the case when the processor is already performing other tasks, such as control of acquisition or communications.

Lastly, the biosignal can be displayed to physicians and users, as occurs in common hospital instrumentation (e.g., intensive care monitors) or, in wireless health monitoring systems, sent wirelessly to a remote location either in real time or offline, to allow observation and monitoring.

As described above, signal acquisition appears to be a straightforward procedure. However, many problems can be encountered when performing data acquisition from biomedical devices for use in clinical applications. Many of these problems are due to the fact that the subject under observation is moving, particularly when considering wireless devices. This causes changes in the measurement conditions. These measurement conditions include the registering positioning of the transducer and changes in body impedance due to movement, chest movement due to arm movement, as well as breathing. For clinical applications, data

acquisition may often be conditioned by the requirement that the measurement device must be applied by a caregiver, and not by someone without medical training. Otherwise the signals have no clinical significance.

Signal acquisition may also be conditioned by requirements from the medical side and not from the signal side. In this way, it is a practice to show the acquired signal to a medical doctor and check if the signal *is good enough for diagnostics* and not look at the time and spectral content of the signal and decide all the details based on that. It is difficult to find specifications for the desired characteristics of signal acquisition. For many biosignals, it is therefore hard to specify the sampling frequency, number of bits for digitalization, resolution, signal latency, delay between signals, etc.

74.3.3 System Integration

When developing electronic devices to give mobility to patients, it may be attempted to achieve different degrees of mobility. A system may allow the patient to move inside a clinical facility. In this case, the device must be designed to be portable, so the patient can move around the facility. Alternatively, the system may allow the patient to be monitored at home. This is similar to the previous case, but with additional requirements. Such a system must have a set of alarms to inform the patient that something is not right (not only in terms of their medical condition, but also that something is wrong with the equipment itself or its use). Moreover, since the user may be required to place the equipment himself, it should be planned in such a way that the patient is able to place it without the need for caregiver assistance, otherwise the equipment will not be used or will be used improperly. Also, since the equipment may be required to stay at the patient's home for some time, desirably, it must be designed so that it can be cleaned at his home. Also, it must be more resistant to potential crash accidents. Another approach is to design the electronic device so that the patient can wear it as if it were just another piece of clothing.

Regarding wearable devices, a few different approaches may be adopted. One solution is based on discrete modules that are placed at different locations where signals need to be measured. However, for some applications, this may not be desirable; for example, to record an ECG, it is necessary to have physical connections between the different leads and the instrument's amplifier, and this solution implies the use of a few wires on the patient body. Moreover, with this solution,

it is difficult to clean and control correct placement of the different measuring devices. The other solution is to design the sensors to be integrated with clothes, with the electronics to be attached to those sensors at a conveniently location. This implies integration of wires with textiles and the development of interconnection technologies to interface wires and electronic devices on textiles. Another approach is to embed all the electronics, wires, and sensors on a polymeric foil, leading to a smart sheet of material with the ability to record the desired signals. Still using the integration approach, instead of electric methods, optical fibers with optical sensors may be integrated, thereby reducing the problem of integrating wires and electronics [74.56].

For wearable devices, the system should be developed taking into account that the user must be able to apply it himself. No help will be provided from any health professional. In this way, it must be developed in such a way that, after putting it on as any regular piece of clothing, the system should only require intuitive adjustments to achieve correct operation. Moreover, since it is something to wear, it must be able to undergo the same cleaning processes as regular clothing.

74.3.4 Wireless Communications

As in many other scenarios, wireless communication is a desirable solution for sensors used for data acquisition in biomedical applications. It is possible to distinguish three main situations: communication between devices inside the human body (intrabody), communication between devices on the body, and communications between devices on the body and the outside world (body to air).

A few aspects may be considered when implementing wireless communication for these different situations. For devices inside the human body, the available power of those devices is very limited, it is very difficult to recharge them from outside, and RF losses

are very high since the wireless medium is highly conductive. This implies that the link must be made at low frequencies (kHz to MHz). This is a problem because, at these frequencies, antennas tend to be very large or, if their size is reduced to fit inside the devices, antenna efficiency drops.

For devices on the body, one main characteristic is operation of the device very close to a highly conductive medium. This has implications for antenna design and signal propagation. Antennas must be designed so that the signal propagates close to the human body, otherwise power is wasted, and it may be required to operate in the antenna near field, as opposed to the standard operation in the far field. It is difficult to predict where devices must be placed to obtain an efficient wireless link.

In the latter two scenarios, a module with limited power may be used if easily rechargeable. The main challenge is to hide the antenna on the user environment while maintaining device operation. Solutions such as placing monopole antennas on the user's hat, which may work, are not desired since they are not fashionable. Many small modules are already sold with small antennas on them and seem to be very convenient for direct use. However, these antennas are designed to operate in free air, and when they are placed close to the human body, their operating frequency and bandwidth, as well their radiation diagram, will be modified.

When dealing with wireless communication, one main problem is to guarantee that data are delivered with sufficient quality of service (QoS) for the desired purpose. It is already possible to find many devices that transmit the measured data wirelessly, but it is very hard to find one that can be used for clinical applications. One problem is the lack of solutions to guarantee QoS for biomedical wireless links. Since data are to be used for clinical decisions, they must be obtained through solutions that guarantee that acquired data are correct and complete, otherwise patient health may be at risk.

74.4 Power Supply

All electronics and sensors require a power supply for operation. Electronic devices in medicine must have a very small form factor, including the power supply. In this regard, it is necessary to use batteries with very high charge density. However, despite all the solutions available, there is always a need to recharge the batteries.

Batteries for wireless wearable devices may operate for 1 day or for 1 week, but, as for cellular phones, recharging will always be required for system operation.

Several wireless AAL solutions for monitoring have been proposed in the past few years, e.g., [74.48, 49]. Most of them are based on standards for communica-

tions on wireless local-area and sensor networks, such as Wi-Fi (802.11), Bluetooth (802.15.1) and ZigBee (802.15.4), because of the availability of commercial modules that can be customized for any wireless application. However, a few solutions that make use of other unlicensed industrial, scientific, and medical (ISM) bands exist. Thorough reviews on wireless sensor networks and related standards can be found in [74.17,57].

Since wireless monitoring is required, those systems must operate on battery power, or they must use an energy harvesting scheme [74.58]. Either way, system power consumption must be kept as low as possible. This leads to the design of systems where power consumption must be optimized. This is done by selecting the most power-efficient components available. If energy is consumed carelessly, the wireless acquisition system may rapidly become completely useless due to lack of available power. To prevent such failure, energy should be carefully saved using different approaches; for example, if the patient is in the normal state, then the sampling rate of sensors can be reduced to save power, or if the battery charge becomes low then its energy should be reserved for more vital tasks; i.e., the monitoring activity should adapt in accordance with the patient's clinical state to save energy. It should be realized that, to save further energy, communication protocols should be simple, and data should be aggregated, compressed, and transmitted in loaded packets, since computation demands much less energy than transmission. However, attention must be paid to the delay, which tends to increase linearly with packet length.

74.4.1 Battery Power Budget Considerations

Since the battery is the main power source for wireless devices, it should be selected to fit the requirements of the electronics which it will supply. The main requirement is its ability to power the electronic device for as long as possible, always considering a minimum period that is acceptable for operation. In many applications this minimum period should not be less than 1 day. After battery autonomy, and many times as important, come the issues of battery dimensions, format, and conformability.

However, other issues must also be taken into account when performing this selection. One is the recharging time, which must be appropriate for the application, since many devices use embedded batteries, which are difficult to replace. This time is usually ac-

ceptable if it is below 3 h, the time that the device may not be used. Also, due to battery lifetime, often it is convenient to use a battery with a protection circuit, which prevents the voltage from dropping below a safe value. In this way, the recharging is faster, at lower currents, and the device will have increased lifetime. Another parameter that is used for battery selection is its nominal current. However, its ability for peak currents also must be taken into account. Higher values of currents may be required, e.g., when a module has wireless communications. When the wireless link is in transmission mode, peak currents may be required. Finally, the operating temperature also interferes with battery lifetime.

Notwithstanding attempts to follow the previous guidelines, battery selection may be a hard task to accomplish, first because often it may not be easy to know beforehand the real power consumption of the device. Power consumption depends, e.g., on the data rate, operating mode (sleep, standby, transmitting, receiving), error rate, antenna efficiency due to antenna positioning, and power level of transmission [74.59]. This implies that tests have to be carried out under real scenarios, or that the battery has to be overspecified.

Another problem is that it is only possible to determine the power consumption accurately when the system is already assembled and ready for operation, where predicted values of system autonomy may diverge from those expected [74.59]. This may be due to several factors, including incomplete manufacturer specifications, only standalone power consumption being taken into account, and power consumption dependence on sampling rates, clock frequency, etc.

Figure 74.6 shows three modules (Wi-Fi, ZigBee, and proprietary) designed with off-the-shelf components that were used for power autonomy testing purposes. The measured current value for the Wi-Fi module, during transmission, exceeds that expected by about 60 mA. Also, in sleep mode, the measured current was 1.5 mA, far above the expected value of 6 μ A.

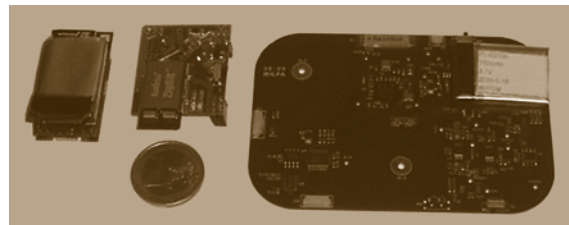


Fig. 74.6 Wireless modules used for autonomy testing: Wi-Fi (left), ZigBee (center), and proprietary (right)

The measured values of the ZigBee module were similar to the values specified in the datasheet. The ZigBee module presented higher consumption than the proprietary module but lower than the Wi-Fi module. The measured current consumption of the proprietary module was slightly higher than the value specified in the product information. The proprietary module had the lowest power consumption, and the measured power consumption was close to the predicted value.

From the power measurements, it is also evident that it may be worthless to design extremely low-power acquisition electronics, sometimes sacrificing acquisition performance, when the main power consumption may be due to the wireless subsystem that, even for the lower-power solution, is one order of magnitude higher.

Finally, it was verified that all the devices were able to work on a very small battery for periods of 1–3 days, which we believe is acceptable for AAL solutions. The Wi-Fi device, as expected, was more power hungry; however, it transmits at much higher power levels, and the data rate is also much higher. Since all microcontroller units have similar power consumption, it is expected that, if the RF power consumption is reduced, the overall Wi-Fi system consumption of this device may be significantly reduced.

74.4.2 Wireless Power

Most medical electronic devices requiring the use of batteries will need to be recharged. The most desirable solution would be one where the user does not need to pay too much attention to its use. The easiest approach is to use a wireless power link, and even more interesting would be that the device may be recharged without taking it out. This is desirable because it is uncomfortable to remove and replace the device, or because continuous monitoring is required. Another benefit of such solutions is power cable removal, allowing for easy fulfillment of requirements such as devices without connectors or that are completely sealed. Moreover, it allows the development of innovative solutions, such as recharging devices that are inside the body, or to recharge devices as we sleep. For short distances, two solutions may be used for wireless recharging: capacitive or inductive. It is also possible to use microwave links or optical links, or other forms of energy, but these options lie outside the scope of this chapter.

From basic transformer theory, we know that the voltage and current available at the receiver depend on the voltage and current at the transmitter. The inductive solution for wireless power transmission uses

two inductors: the primary and the secondary. On the transmission side, a power amplifier is used to drive the primary inductor. This power amplifier must be matched to the inductor in order for the available power to be transmitted to the inductor and the maximum power to be obtained in the secondary. The inductor acting as a receiver at the secondary side is connected to a power rectifier, after which comes a circuit to control the battery charge. To obtain maximum efficiency, the receiving inductor should also be matched to the load. However, as the battery charges, the current usually decreases and the load presented to the inductor also changes. This implies that, for continuous matching, the matching circuit should change during the charging period.

74.4.3 Wireless Power Link Analysis

To understand the losses expected from this antenna solution, a wireless link was implemented and analyzed. An external (transmitter) subsystem consisting of a Collpitts oscillator was used to generate an AC signal to feed the power amplifier. The implanted (receiver) subsystem has an AC–DC converter for power recharging purposes, and the data link path is preceded by a notch filter for power link rejection [74.60]. In this way, it is possible to power the receiver as well as to receive data transmitted wirelessly. The wireless link was implemented using two coils in parallel with the following dimensions: width, 2 cm; length, 4 cm; thickness, 0.254 mm; number of turns, 18; distance between each turn, 0.254 mm (Fig. 74.7).

As the matching network for the inductors, four capacitors were used, two in series (4.7 nF) and the other two in parallel (2.2 nF), in order to avoid losses due to power mismatch in the wireless link. The match-

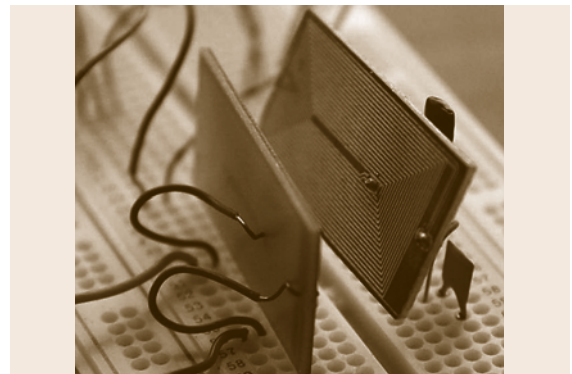


Fig. 74.7 Setup based on the inductive link antenna

ing network was trimmed with the help of simulation to optimize the efficiency of the wireless link. The wireless link was designed to operate at a distance of 3 mm between the two inductive coils. To investigate the link performance, measurements, simulations, and analytical models were used. Measurements were made for the inductive wireless link and compared with simulation results. The link efficiency was computed and then compared with the efficiency obtained from analytical models for other antenna solutions. Return loss measurements were performed using a vector network analyzer, enabling efficiency measurements. On the other hand, all simulations were done using Ansoft Designer SV software.

First, the inductive wireless link was characterized without any matching network and compared with the results obtained from simulations. The results showed high losses, but were made with the purpose of providing a way to design the matching network as well as to validate the simulations performed previously. Using the measured results as an input for the simulator, it was possible to design the corresponding matching network and compute the capacitors required for matching. Figure 74.8 shows the result of that matching.

It can be observed that the value of S_{11} was below -10 dB, meaning that the power is being injected, and S_{21} was -3.90 dB, meaning that 65% of the energy is not being transmitted.

After designing the matching network, the wireless link was measured again. The measured S -parameters are shown in Fig. 74.9. The efficiency measured for the

inductive wireless link with real capacitors was not as high as that obtained theoretically, because of capacitor tolerances and quality factors. So, for 1 MHz, the value of S_{11} was -19.25 dB, which means that energy is entering port 1 (the transmitter), although the value of S_{21} is below the theoretical value (-5.02 dB), meaning that more energy is being lost (83.7%).

To test the effect that neural tissue may have on the performance of the system when used to recharge neuronal biodevices wirelessly, the validated model was used for testing the case when a slice of human tissue with thickness of 3 mm was placed between the two coils. Figure 74.10 shows the simulation results for such an inductive wireless link.

These results were not as good as the results shown in Fig. 74.8 for air, due to signal attenuation caused by the tissue. In this case, the dielectric properties were selected to match the neurological tissue behavior. The results of this analysis show that energy was being injected, because the value of S_{11} was simulated and tuned to be below -10 dB (-13.33 dB). On the other hand, as illustrated in Fig. 74.10, S_{21} was -4.51 dB, which means that 75% of the energy was being lost, which includes antenna and tissue losses.

74.4.4 Wireless Power Charger

Despite the losses expected when using wireless power transmission, there are situations where implementation of this option is required [74.61]. Figure 74.11 shows a prototype of a system developed to recharge

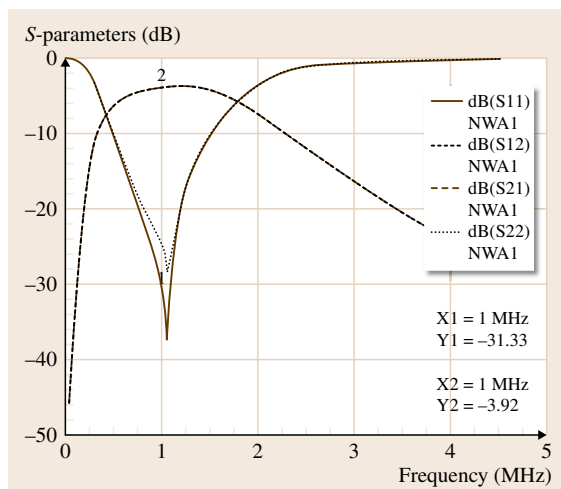


Fig. 74.8 Measured S -parameters and matching with ideal capacitors

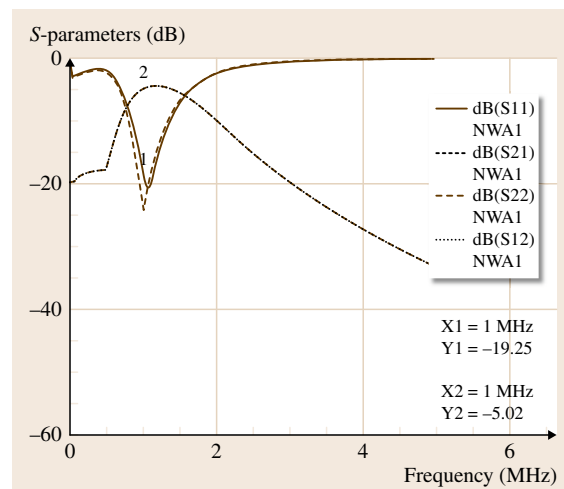


Fig. 74.9 Measured S -parameters for the full wireless link

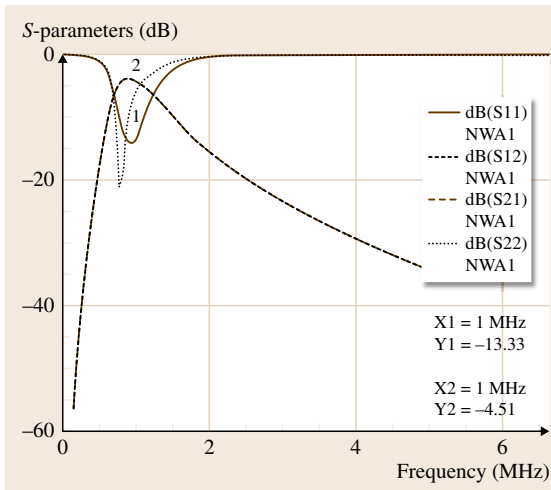


Fig. 74.10 Simulated *S*-parameters, with neurological tissue’s properties

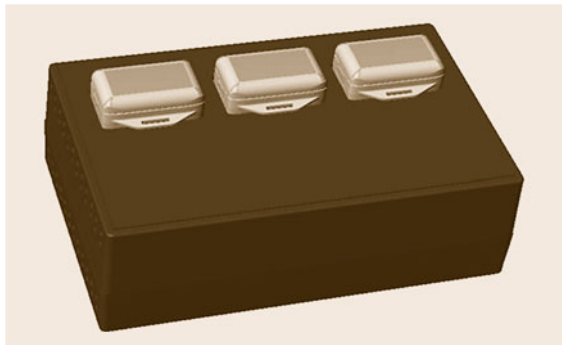


Fig. 74.11 Wireless charger device



Fig. 74.12 Wireless power (power mattress and power pillow)

three wireless modules used for AAL applications. With the developed wireless link, it is possible to recharge the three modules in about 2.5 h. The modules have autonomy of approximately 20 h for their planned application.

The solution presented in Fig. 74.11, despite being effective, requires removal of the wearable device for recharging. A desirable solution would have the ability to recharge such devices when the user is performing some activity, close to a location where it is possible to place the charger. One solution that may be adopted is the use of a charger format that may be transformed into a *power pillow* or a *power mattress* (Fig. 74.12).

Another solution is to place chargers on equipment at the gymnasium, or on a desk or desk chair.

The idea behind these concepts is to provide a solution to recharge the wearable devices without removing them. Using this concept, the power is transmitted to the system’s battery while the user is lying in bed, either to sleep or when at the hospital for diagnostics or treatment. However, these kinds of solutions, beyond the technical problems that must be solved before their implementation, require also deeper knowledge of the effect that the recharging electromagnetic field may have on user health.

Despite the many benefits envisaged by this technology, it still has a long way to go before it can be widely deployed. This technology suffers from a few problems, which are hard to solve, with link efficiency being one of the most important [74.62, 63]. To achieve efficiencies on the order of 70–80%, very special conditions must be met. Regular links will not reach above 50%, even for short transmission distances. However, this solution may be very interesting for applications where the charging device and wireless module may be placed close together.

74.4.5 Energy Harvesting

Another option available to power wireless devices is the use of some form of energy harvesting. Any form is valid, since it is possible to use that energy to charge a supercapacitor or a battery. Many forms of energy conversion may be used to benefit from the available energy sources: thermal, electric, mechanical or chemical [74.58]. Despite the many proposed solutions, they still cannot provide sufficient power to drive complete medical electronic devices continuously. In fact, it is already possible to find devices operating on this kind of power sources. However, they tend to work in very

specific conditions and for very specific situations. The amount of power that can be obtained from these solutions is still below the requirements [74.58]. Moreover, they require very advanced electronic solutions. All the required electronics must be based on very low-power

designs, and wireless transmission must occur at very low-power levels.

However, and notwithstanding these limitations, this may be a potential solution as an extra power source to help increase battery lifetime.

74.5 Wearable Medical Electronics

Electronics in medicine may be used on fixed devices, portable devices or wearable devices. In this section, electronics in medicine will be addressed from the perspective of wearable devices.

Wearable technology represents a new, emerging field with increasing potential in several aspects of modern healthcare, with greater importance at the delivery point of care services. As a result, remote and ambulatory monitoring inside or outside of the healthcare environment, and support for disabled, rehabilitating or chronically ill patients, is now a reality. The growing demand for wearable devices is being driven by the considerable need for preventive instead of reactive medicine, the global increase of health awareness, and the need for proactive personal healthcare on a daily basis.

Figure 74.13 show examples of the most commonly monitored biosignals on the human body [74.44–46].

One aspect that may be highlighted is that many important biosignals are electric signals, therefore they can be measured with similar instrumentation. On the other hand, different technologies must be available to

monitor all the required signals, since they have different origins. Another aspect is that, to measure all the required signals, sensing devices must be placed in different body locations, which requires the development of specific solutions.

A wearable medical device can be described as an unobtrusive, autonomous, and ubiquitous system that supports continuous, multiparameter monitoring and treatment, and telemetric abilities [74.64, 65]. This contributes to a shift of health services from conventional hospital-centered towards individual-centered healthcare, which together with wireless technologies allows continuous feedback of relevant information to the user and/or clinical professionals. In addition, this approach improves early detection of and timely response to possible health threats [74.64, 66].

Since they are wearable, these devices are truly portable, being supported either directly on the human body or in a piece of clothing. This results in a new set of requirements, including minimal size, light weight, portability, unobtrusiveness, high levels of flexibility and integration, ease of use and comfortable wearing, increased lifecycle, and power autonomy. Such requirements can now be fulfilled, given the trends towards miniaturization of electrical and electronic equipment, as well as novel integration and sensing techniques. In particular, advances in sensing technologies allow monitoring and processing of two types of signals: vital signs such as cardiac activity, blood pressure, and temperature; and other important parameters that are more related to the patient's condition, such as body kinematics, sensorial and cognitive reactivity, and others [74.64, 65].

Due to the overall results of the advances in both the technology and healthcare sectors, it has become possible to establish a new healthcare paradigm: personalized health systems. These will enable decentralization of healthcare towards a system that will give the user a more proactive role in their care, providing better monitoring and feedback through a comfortable and unobtrusive solution. This not only contributes to im-

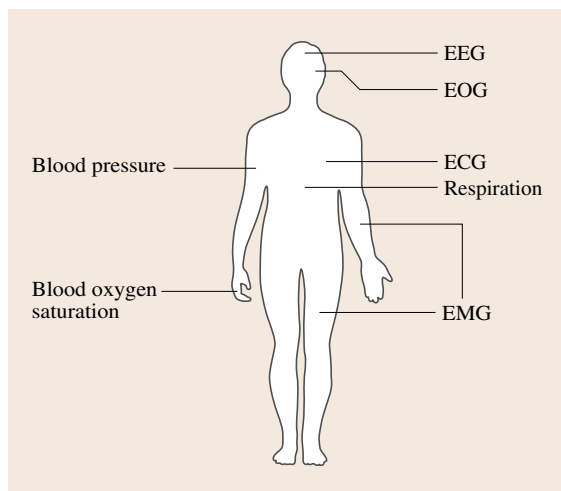


Fig. 74.13 Examples of noninvasively measurable biosignals on the surface of the human body

proved quality of life, but also allows for preventive care by effective diagnosis at the earliest signs of disease. Likely to be of benefit to the chronically ill and disabled, wearable health devices are an attractive solution for patients undergoing rehabilitation, providing them



Fig. 74.14 SensatexSmart Shirt (Sensatex, Inc., Bethesda, MD)



Fig. 74.15 Lifeshirt from Vivometrics (Vivometrics, Ventura, CA)



Fig. 74.16 BodyMedia arm band (BodyMedia, Pittsburgh, PA)

with independent living, since they enable recording and collection of relevant data in the different situations of individual daily life [74.65,67].

A number of wearable devices in the healthcare sector have emerged in the past few years, ranging from simple monitoring of daily routine, to miniaturization and integration of sensors to enhance the overall performance of wearable systems. The Georgia Institute of Technology (Atlanta) jointly with the US Navy proposed one of the first wearable solutions, consisting of a wearable vest embedded with optical fibers and sensors, working also as a data bus – the Georgia Tech Wearable Motherboard (GTWM) [74.68]. All the components are integrated into the fabric, creating a flexible device, which was manufactured essentially for use under combat conditions. The GTWM has the ability to identify the exact location of injuries or physical problems, helping to determine which of them needs immediate attention. This device was placed on the market by Sensatex, Inc., as a product named the Sensatex Smart Shirt (Fig. 74.14).

Companies such as Vivometrics and Bodymedia have also entered the market with wearable monitoring systems. The Lifeshirt (Fig. 74.15) is a product by Vivometrics and consists of a wearable physiological monitor in the form of a chest and shoulder strap, providing noninvasive ambulatory monitoring of pulmonary cardiac function and posture [74.69].

Bodymedia introduced a wearable and wireless body monitor that continuously collects physiological and lifestyle data such as heart rate, heart flux, body and ambient temperature, and galvanic skin response, among others. The Bodymedia Fit consists of an arm-band (Fig. 74.16), a wirelessly connected display, and an activity manager [74.70].

Many research groups have started to develop wearable technologies with main applications in health science. In the work entitled *AMON: A Wearable Medical Computer for High Risk Patients*, a wearable medical monitoring computer is presented [74.71]. The AMON system was developed under a European-sponsored consortium and consists of a wrist-worn unit with monitoring, data analysis, and communication capabilities. This system is mainly intended for high-risk patients in need of constant monitoring. The paper by *Paradiso and De Rossi* describes a project funded by the European Commission jointly with Philips and other partners, both industrial and academic [74.72]. This project, called *MyHeart*, consists of the development of functional clothes with on-body sensors and electronics to acquire, process, and evaluate physiological

data. *Choi* and *Jiang* have developed a wearable sensor device in the form of a belt-type sensor head, which is composed of conductive fabric and polyvinylidene fluoride (PVDF) film, for monitoring cardiorespiratory signals during sleep [74.73]. *Pandian* and coworkers [74.74] have presented a wearable vest – the Smart vest – with integrated sensors for monitoring of ECG, photoplethysmogram (PPG), body temperature, blood pressure, galvanic skin response (GSR), and heart rate. In a more recent work, *Kanellos* and his coworkers developed a flexible optical-fiber-based pressure sensor comprising fiber-embedded Bragg grating elements [74.75].

The aim of this section is to introduce the enabling technologies behind the design of wearable devices in healthcare, as well as the basic requirements for their design.

74.5.1 Wearable Systems

The ongoing miniaturization of computing hardware and electronics as well as new smart materials and integration techniques have led to the development of noninvasive sensors and actuators for monitoring physiological and kinesiological parameters. These systems can include various types of small physiological sensors, transmission modules, and processing capabilities, facilitating low-cost wearable unobtrusive solutions for continuous all-day, any-place health, mental, and activity status monitoring.

A wearable system can be described as a smart/functional material with embedded sensors, processing and communication units, which may or may not contain attachable peripherals or other appliances. One can refer to this type of system as wearable electronics.

To design a wearable system, three areas of work need to be properly covered. First, it is important to develop unobtrusive wearable sensors to record physiological and kinesiological data reliably. Secondly, these sensors need to be implemented into a substrate material that allows multisensor integration. Finally, it is important to provide the system with acquisition and communication infrastructures, to enhance mobility of individuals, and some degree of intelligence to improve system performance at a clinical level [74.64, 65].

Figure 74.17 depicts an architectural layer for an ideal wearable health system, which is composed of a functional/smart substrate, embedded electronics, and attachable peripherals/appliances.

Wearable system components may be divided into three main categories: clothing, embedded electronics,

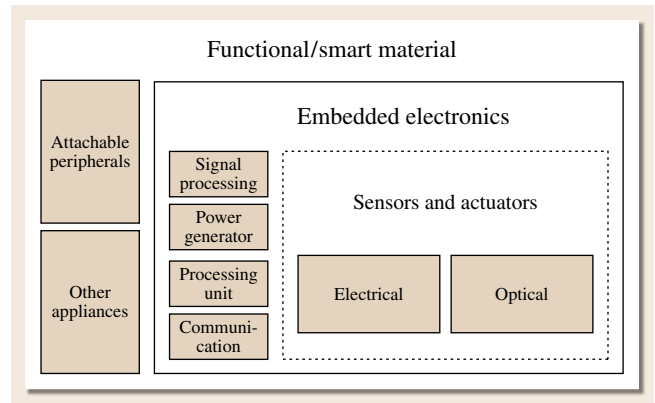


Fig. 74.17 Architectural layer of an ideal wearable health system

and attachable components. The first includes all the substrate materials that act as a functional or smart structure by allowing embedding of sensors, signal processing units, communication infrastructures, actuating mechanism, and power generation, among others. The most common substrate materials used in these systems are textiles or flexible polymeric materials that will enable the design of garments with multifunctional nature [74.76]. The embedded components include all the necessary electronics, optics, or other devices that will provide sensing, actuating, signal processing, communication infrastructures, power generation, and other necessary functions. If necessary and desirable, attachable peripherals and other appliances can be included into the wearable system. Examples of these types of components are personal digital assistants (PDAs), displays, keyboard, and control knobs, among others.

Focusing on the sensing technologies, new approaches are required, especially to provide noncontact methodologies to improve embedding of these components as well as to contribute to the design of totally wearable and highly comfortable functional garments. Sensors are used to monitor all the necessary physiological parameters and the physical environment surrounding the user, allowing the user's health condition to be monitored. They can either be embedded in the material, or the material itself works as a sensing element. These components will be further discussed below.

74.5.2 Categories of Wearable Systems

Wearable systems can be classified according to the level of integration of the components into the smart/functional material, i.e., the substrate. There are

three types of wearable systems according to this classification: first generation, based on attached hardware components and sensors; second generation, where these components are embedded; and third generation, where innovative integration techniques during the substrate material production allow for the design of multisensory clothing and/or accessories.

The first category refers to all devices based on plug-in/out methodologies, where a supporting mechanism for attaching the necessary components is provided. These can include ECG monitoring wristwatches, and sensing components that can be attached to a t-shirt, a vest or even a cap. The problem associated with these types of wearable devices is their lack of comfort, and they are not yet a practical solution from the user perspective. In addition, since they are based on attached components that usually are not completely hidden, these systems generally do not allow for a completely unnoticeable solution.

The drawbacks of the first generation of wearable systems led to the creation of a new generation based on partially integrated/embedded architectures, where all the necessary components are fixed to the substrate material. This not only eliminates the need for qualified personnel or for the user to place the components, running the risk of misplacement, but also allows for a more practical and discrete solution. However, there is still a considerable difference from a normal garment or accessories, meaning that the components do not have a sufficient level of integration into the substrate, providing relatively comfortable solutions but still remaining perceptible.

Research and progress in integration techniques during the fabrication process allowed for the design of a third generation of wearable health devices. This type of system represents the cutting edge of wearable technology, allowing the design of smart, functional, and multisensing materials, which are apparently normal, due to this high level of integration. A very popular example of a third-generation wearable system is the electronic textile e-Textile, which consists of high-knowledge-content garments made from multifunctional fabrics. By blending the components into the user's ordinary clothing, it is possible to achieve an ideal wearable system, minimizing the hassle of wearing the device.

Despite the existence of these three generations of wearable devices, with their drawbacks and advantages, they all provide enhancement of user quality of life, allowing for real-time multiparameter monitoring under various conditions.

74.5.3 Design Requirements

The design of wearable systems requires an additional set of requirements, especially when compared with stationary equipment. If the final goal is to provide a customized multisensing wearable system that can be worn during daily life as normally as before, it is imperative that the wearable system cover the following set of key points [74.64, 65]:

Interaction

The device should be able to interact with the environment through a network of sensors placed in different parts of the clothing or accessories, allowing the creation of a certain awareness of the physiological and emotional state of the user, as well as of the surrounding environment. This is extremely important when designing smart and functional devices that are required to monitor, control, manipulate, interact, and actuate on a real-time basis. Data handling, decision support, and feedback are also crucial to establish a good interaction between the device, the components, and the user himself.

Autonomy, Safety, and Reliability

This aspect deals with the power and operational requirements which have to be met, without which autonomous, safe, and reliable operation of the device would be impossible.

Easy User Interface

It is important to maintain an easy interface between the user and the wearable device, minimizing the user's cognitive effort and involvement in the process.

Integration

It is crucial to improve integration of the components into the substrate in order to enable normal clothing and accessories with multifunctional capabilities. In addition, it is also important to interfere as little as possible with the user's physical activity, which is achieved by full integration of components into the substrate material.

Along with these key points come the user and design requirements for wearable systems. From the user's perspective, the most important features are minimal size and light weight, unobtrusive interface, user-friendliness, comfortable wear, ergonomics, portability, and increased lifecycle. When considering other perspectives in terms of design and performance, in addition to taking into account the user's requirements,

there are a few key requirements to consider. Power autonomy, housing conditions, wireless communication, continuous time monitoring, and high levels of flexibility and integration are the main requirements from the design and performance perspectives.

By fulfilling the above requirements it is possible to design a wearable system that allows the user to benefit from its multifunctional and multisensing capabilities, without compromising the user's daily life activities.

74.5.4 Sensors for Wearable Systems

In wearable devices, a wide range of sensors are used to measure physiological and environmental conditions. The first type of sensors – biosensors – are used to monitor a clinical condition or process, therefore measuring biopotentials and other physiological parameters. On the other hand, the second type of sensors – peripheral sensors – are responsible for sensing the surrounding environmental conditions, enhancing the awareness of the system.

Not every sensor can be used in a wearable context, especially from the user's perspective. Not only do their physical attributes, such as size and weight, have to be taken into account, but also their noninvasive character and ease of placement must be considered. In addition, these sensors must ideally produce an electrical output in order to be digitally processed, being durable and reliable and with low power consumption [74.64, 67].

Usually, sensors used in wearable devices are based on conventional sensing elements, such as piezoelectric materials, electrodes, and others, with integrated communication infrastructure (e.g., wireless communication). This eliminates the problems associated with cable management and allows sensors to be positioned

at various locations without any complications. In an ideal system, sensors should not require any contact with the user, since they could benefit from integration into clothing, which from the user's perspective contributes to the design of more practical, comfortable, and easy-to-use devices.

74.5.5 Sensing Methodologies

A number of technologies can be applied as sensing methodologies, such as electrical, mechanical, optical, piezoelectric, among others (Table 74.2) [74.12, 77, 78]. However, it is important to consider the demands associated with sensor integration into textiles or other flexible materials.

Sensors are composed of materials that respond to a physical stimulus, such as a change in temperature, pressure, electric field or illumination. They can range from skin surface electrodes to conductive yarns, providing comfortable and practical solutions for the user. Several materials can be used as the sensing component, and they can be divided into different categories, each with a specific effect.

The general acquisition system of a typical sensor used in wearable devices is composed of several functional blocks, from signal acquisition to output of the final result. Figure 74.18 shows a generic acquisition block diagram for a wearable sensor.

The transducer converts the desired signal or variable into another form of energy, which is then amplified and filtered to remove undesired signal components. The obtained analog signal is converted to digital form by an ADC and further processed using many different algorithms. If necessary, the processed signals can be converted back to analog form to drive

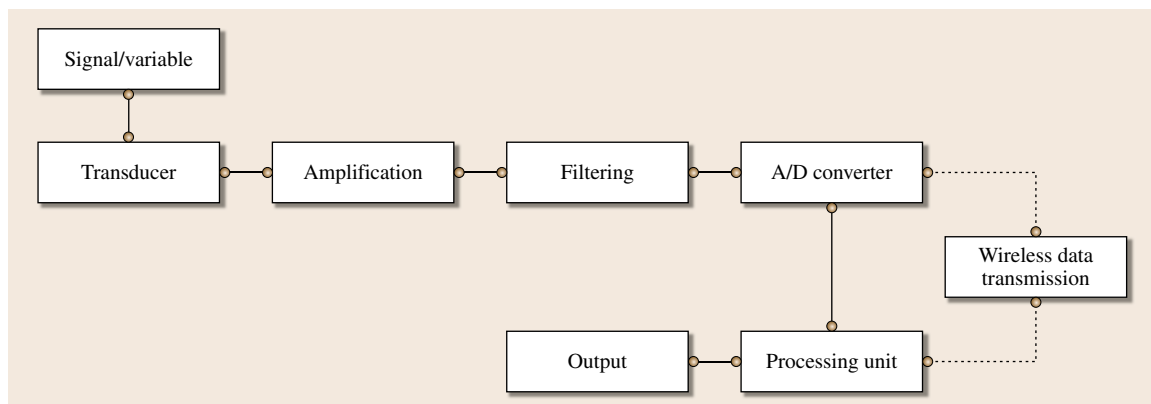


Fig. 74.18 Signal acquisition block diagram of a generic wearable sensor

Table 74.2 Different transducer effects applied in the sensing mechanism of wearable devices

Transducer effect	Sensing devices	Stimulus	Response	Examples of materials	Examples of detected signals
Electrochemical	Bioelectrodes, electrochemical cells	Biochemical activity	Electrical current	Copper, aluminum, Ag/AgCl, stainless steel, conductive yarns, electroactive polymers (EAPs)	EEG, ECG, EMG, EOG, skin electrode impedance
Electrooptic	Electrooptic modulators	Electric field	Birefringence	Lithium niobate (LiNbO ₃), lithium tantalite (LiTaO ₃), deuterated potassium dihydrogen phosphate (KD ₂ PO ₄), electrooptic polymers	EEG, ECG, EMG, EOG
Electrostriction	Electrostrictor transducers Example: pressure sensor	Direct effect (nonlinear) Mechanical stress Reverse effect (nonlinear) Electric field	Electric polarization Dimensional change	Lead magnesium niobate, lead lanthanum zirconate titanate (PLZT), EAPs	Carotid pulse, heart apex pulse, ECG, movement
Electroluminescence	Light emission devices	Electric field	Light emission	EAPs	ECG, EMG
Photoluminescence	Photoluminescence sensors Example: UV radiation sensor	Incident light	Light emission	EAPs	UV radiation
Piezoelectricity	Piezoelectric transducers Example: strain gages	Direct effect (linear) Mechanical stress Reverse effect (linear) Electric field	Electric polarization Dimensional change	Polyvinylidene fluoride (PVDF), barium titanate (BaTiO ₃), lithium niobate (LiNbO ₃), lead zirconate titanate (PZT)-based compounds, EAPs	Carotid pulse, heart apex pulse, ECG [Samjin Choi]
Pyroelectricity	Pyroelectric sensors Example: infrared sensors	Temperature change	Electric polarization	All pyroelectrics are piezoelectrics	Body temperature, ambient temperature

specific devices. For better design, it is often necessary to include wireless communication systems that enable the sensing component to transmit the data to a control processing unit or even to a database service.

74.6 Electronics in Medicine at Work

After introducing several aspects of medical electronics and several signals that may be recorded by such systems, this section presents a few implementations of electronic systems designed for medical applications.

74.6.1 Wireless and Wearable Low-Power Health Monitoring Systems

A prototype wearable, low-power acquisition module for a WHMS is shown in Fig. 74.19. The acquisition module provides continuous monitoring of single-lead ECG, temperature, and activity [74.79]. The module is used on the chest of the monitored subject and can be attached using either a chest band or by integration into a textile garment. In addition, the wearability of the acquisition module is supported by its small size and lightweight design, and by the assembly of all of its electronic components on a flexible polyimide printed circuit board (PCB), which in turn houses dry and flexible electrodes [74.80], directly attached to the PCB without use of wired connections. Unlike conventional silver/silver chloride electrodes, these dry and flexible electrodes are able to acquire a biosignal without the need for skin preparation with electrolyte gel, which is suitable for use in long-term and continuous health monitoring systems.

74.6.2 Sensors

The sensors contained in the acquisition module were chosen considering its continuous monitoring and low-power operation features. The acquisition mod-

ule includes three flexible, dry ECG electrodes with a low-power analog front-end for single-lead ECG monitoring, a negative-temperature-coefficient (NTC) thermistor for surface temperature monitoring, and a low-power three-axis accelerometer for activity monitoring.

The analog front-end designed for ECG monitoring is shown in Fig. 74.20. A pair of the flexible dry electrodes at a short distance from each other (65 mm) is used to acquire a single-lead ECG, when contacting the skin on the chest, close to the heart. The remaining electrode is used as a reference driver connection to reduce common mode interference.

The performance of the analog front-end was designed to achieve a trade-off between low power consumption and suitable common mode rejection ratio (CMRR) by appropriate choice of the instrumentation amplifier (current consumption of 93 μA and CMRR of 115 dB). A notch filter was included in the front-end to attenuate electromagnetic interference from surrounding power sources, since the acquisition module should be able to monitor the subject even in electromagnetically polluted environments. Furthermore, a bandpass filter is used to limit the bandwidth of the acquired signal to the ECG frequency range and to attenuate the DC offset. The complete analog front-end achieved current consumption of about 220 μA .

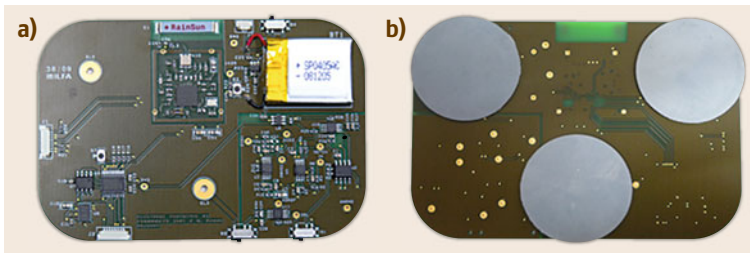


Fig. 74.19 (a) Wearable low-power acquisition module for a WHMS, showing electronics on the top layer and (b) flexible dry electrodes for ECG monitoring on the bottom layer

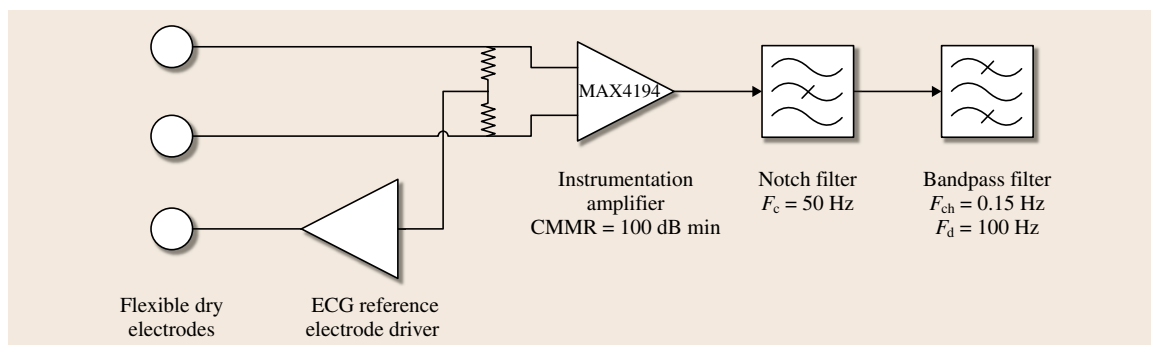


Fig. 74.20 Low-power ECG analog front-end included in the WHMS prototype

The activity of the monitored subject is estimated by the LIS302DL digital accelerometer which has a configurable range of ± 2 g or ± 8 g, featuring low current consumption (300 μ A). The local temperature at the acquisition module can be estimated within the range of 20 and 40 °C, by the change in resistance in the NTC thermistor.

74.6.3 Wireless Link

The wireless communication interface between the acquisition module and a base station [e.g., personal computer (PC) or PDA] is provided by the Toumaz Sensium (Toumaz Technology Ltd., Abingdon [74.81]). The Sensium is a system-on-chip solution containing a low-power transceiver platform, 8051 microcontroller, and reconfigurable sensor interface. The Sensium allows the creation of wireless body-area networks composed of a base station and up to eight target nodes which communicate using a proprietary protocol aimed at minimization of power consumption. The wireless link between Sensium devices operates in the 868/915 MHz ISM band, providing a 50 kbps data rate.

For the design of the presented acquisition module, the use of the Toumaz Sensium solution with a proprietary wireless communication protocol was preferred over the use of standard protocols for wireless sensor networks or wireless local-area networks, such as Bluetooth (802.15.1), ZigBee (802.15.4) or Wi-Fi (802.11). The use of the proprietary protocol has the advantage of low power consumption, achieving longer operation times suitable for continuous monitoring, and avoiding the crowded 2.4 GHz ISM band, thus aiming to achieve better resilience to interference and improved quality of service. On the other hand, use of the proprietary protocol and transceiver reduced transmission power and data rate to achieve low power performance, thus only allowing transmission distances of 10 m and a data rate of 50 kbps. However, this transmission distance may be suitable for monitoring a subject at home or in specific rooms, or if the subject carries a smartphone or PDA as a base station. In addition, the 50 kbps data rate is sufficient for continuous monitoring of most biosignals, considering their limited bandwidth.

74.6.4 Optical Biopotential Recording

In the past few years, optical-based sensors have become increasingly used in healthcare applications. This is explained by the many advantages of these sensors that include immunity to electromagnetic interference,

resistance to harsh environments, freedom from electrical wires, small size, light weight, and multiplexing capabilities [74.82]. Moreover, many approaches for optical fiber integration have been developed, with particular interest for wearable health devices, leading to easier optical fiber integration into textiles and other wearable materials [74.56, 83, 84].

74.6.5 Optical Electrodes

When dealing with optical electrodes, several modifications must be made to the acquisition system shown in Fig. 74.18. A light source and detector must be included, since the sensing methodology is based on alteration of light properties. Figure 74.21 shows a typical acquisition system for an optical sensor.

In this case, the acquisition system must include a light source that will pass through an optical transducer. In the presence of a particular signal, the optical transducer will suffer alterations, for example, a dimensional change. This alteration will produce a shift in light properties, whether intensity, phase, polarization or other. Afterwards, a photodetector is responsible for collecting this modulated light and converting this modification into a desired result. The latter is dependent on the type of photodetector applied in this stage; for example, a photodiode can be used to convert light intensity into current or voltage. The signal is then processed according to the remaining block diagrams, which are similar to those in Fig. 74.18.

An important feature of optical-based sensors is the ability to enable easier contactless measurements of physiological events, particularly electrophysiological signals. This can be achieved using specific transducer effects by which a material exhibits a particular response in the presence of a stimulus such as an external electric field. Materials exhibiting this type of stimulus-response mechanism are classified as electroactive materials; some such materials are included in Table 74.2. The electroactive component works as the sensing element, which can be in the form of a coating material, such as a hydrogel or a piezoelectric material, or even used as a device such as an electrooptic modulator.

For several years, the standard biopotential recording systems consisted of bioelectrodes, instrumentation amplifiers, and filtering components. The signal was recorded by placing electrodes in contact with the skin, allowing conversion of the ionic current within the body into an electronic current in the electrode leads [74.54]. The coupling between the electrode and the skin was

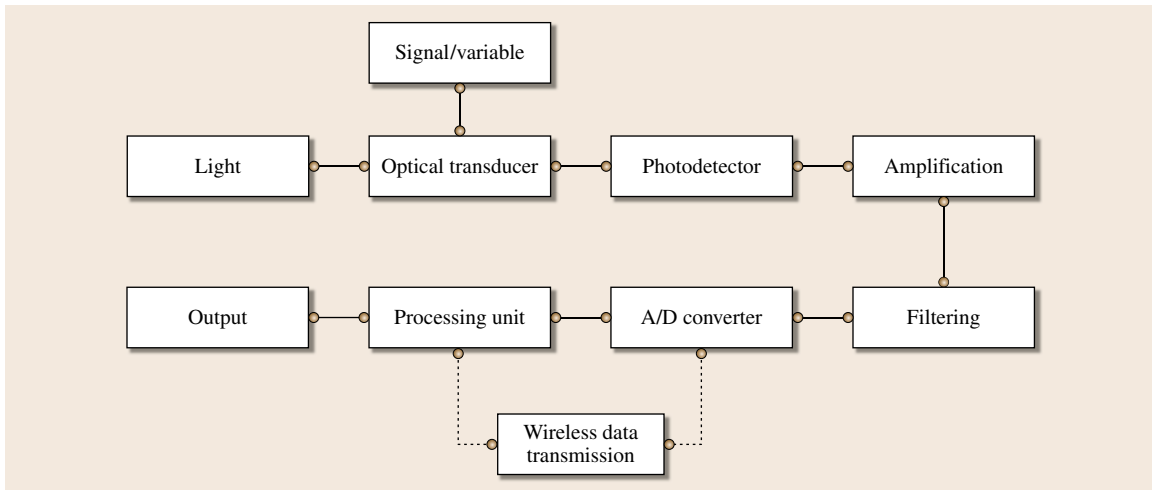


Fig. 74.21 Optical sensor acquisition block diagram

usually made by an electrolyte that could be either a gel/solution for wet electrodes or simply sweat for dry electrodes. This standard recording system, despite its good performance in terms of sensitivity, is not very suitable for wearable applications, since it requires good electrical and/or physical contact with the skin [74.85].

Novel dry and textile-based wearable electrodes have recently been proposed to overcome the limitations associated with such electrodes. These include, for example, conductive rubber electrodes [74.86], Cu-sputtered textile electrodes [74.87], conductive fabric sheets and polyvinyl difluoride (PVDF) film electrodes [74.73], and polymeric dry electrodes [74.88]. However, these still require physical contact with the skin, making optical sensors the strongest candidate for contactless measurements in wearable devices.

In recent work, an electroactive hydrogel was suggested as a transducer for optical-based biopotential measurements for applications in wearable systems [74.89]. This electroactive material, polyacrylamide hydrogel (PAAM), when submitted to an external electric field, undergoes a bending process, altering its mass and volume properties. Likewise, the amount of light passing through the hydrogel and reaching the photodetection stage will change. Although many configurations can be applied in the sensor design, the desirable solution could be a fiber pigtail, with the PAAM hydrogel placed at the tip of the fiber. This material showed an adequate sensitivity and frequency response, as well as the ability to perform better as sample thickness decreases, making it an eligible sensing component for biopotential recording applications.

74.6.6 Optical Signal Acquisition

An example of an optical-based sensor for biopotential acquisition is the electrooptic modulator, which can be used to modulate light (e.g., intensity, polarization or phase) in response to an electric field. The observed modification of the light property is proportional to the biopotential, making it possible to translate this result into a biopotential recording. The electrooptic (EO) sensor proposed is a photonic fiber optic based device, whose main functional stages are optical signal generation, electrooptic modulation, and detection. These EO sensors may be developed and assembled so as to be easily integrated into fabrics or other wearable materials. Figure 74.22 shows an ECG signal obtained using this method as the recording technique.

As shown in this figure, this EO technique has the required frequency response and sensitivity to acquire an ECG signal with its different signal components, such as the QRS complex.

74.6.7 Localization Solutions

One well-known requirement in AAL solutions is the ability to track someone inside an environment using the already deployed wireless infrastructure. Several wireless AAL solutions for monitoring have been proposed in the past few years, e.g., [74.48, 49]. One solution is to use the Wi-Fi infrastructure and deploy small modules, worn by the subject being monitored. Then, using the Wi-Fi network infrastructure, the module

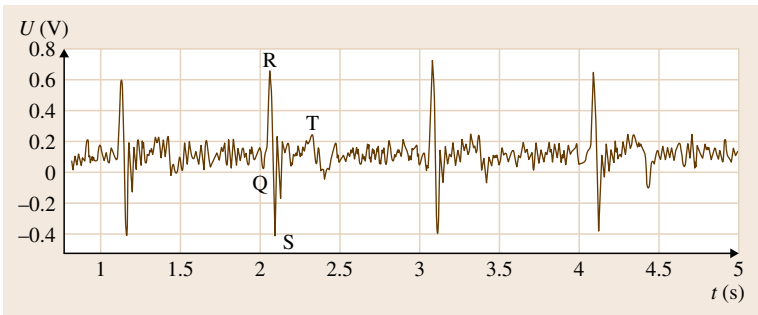


Fig. 74.22 ECG signal recorded with an electrooptic modulator

can be used for tracking purposes [74.48, 49]. Despite the well-known benefits of this technology, its acceptance requires wearable devices, which must be small, lightweight, and with enough power autonomy for comfortable operation [74.61].

Figure 74.23 shows a system that enables localization of a walking user inside a building, using a device that was developed to be used as a wristwatch [74.61]. This is the smallest localization device known for use in localization applications, based on a Wi-Fi infrastructure. This module was designed to fulfill clinical environment compliance, which requires a hermetically sealed module made from clinically compatible materials. Since it is to be worn as a wristwatch, wearability and fashionable design were taken into account. Moreover, for ease of use and cleaning, the module uses wireless recharging technology.

Since the localization device had to be designed to be worn on the subject's wrist, the module should not greatly exceed the dimensions of a wristwatch. Also, and to reduce deployment costs, the localization technology is based on the installed Wi-Fi network, so the localization device must use a Wi-Fi module. The designed module uses an off-the-shelf low-power Wi-Fi tag. The tag communicates with existing wireless lo-

cal area network (WLAN) infrastructure to help control centers to keep track of the subjects being monitored. A key feature of this device is its waterproof capability and the requirement for simple cleaning methods. In this way, a wireless recharging capability was implemented, which enables full recharging of the battery when the module is placed on a base station for around 2 h. Moreover, as onboard sensors, it has a motion detector and a photonic wrist removal detector.

Localization Ability

Figure 74.24 shows the software front-end used in combination with the developed module.

Initial tests showed a resolution of 2–5 m (90% of the time) for an indoor environment. Those tests showed that the tag meets the project requirements. The performance obtained makes the tag viable for all the expected project scenarios, namely in hospitals and healthcare markets.

Module Autonomy

Since the module was expected to be as small as possible, the platforms under test were powered by a 150 mAh, 3.7 V, $20 \times 30 \times 4.0 \text{ mm}^3$ Li-ion battery. However, the indicated supply battery may change, due to different autonomy constraints.

All the tests were carried out using a small resistor in series with the battery, and the voltage drop was measured. From that, the current consumption was extracted. Table 74.3 shows the results obtained when using this procedure.

At the moment, this device is used only for localization purposes. In normal operating mode, the device sleeps for 20 s, then awakes, detects its position, and sends it to a nearby wireless access point.

From Table 74.3, it can be observed that, during transmission, the measured current value exceeds the expected value by about 60 mA. Also, in sleep mode, the current measured was 1.5 mA, far more than the ex-



Fig. 74.23 Magtag, side-by-side with a wristwatch

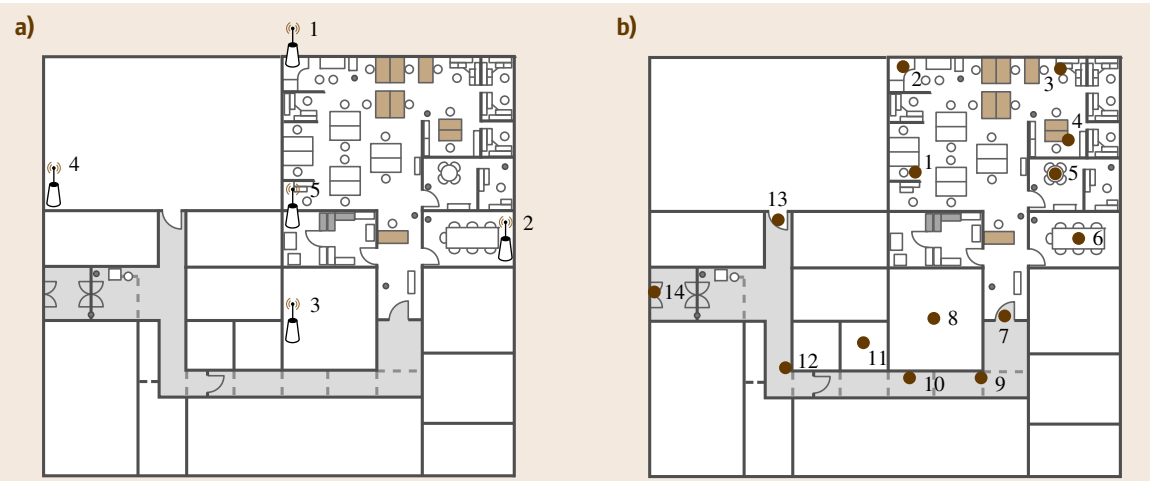


Fig. 74.24a,b Tracking operation. (a) Testbed access point location. (b) Tag positions used for localization error assessment

Table 74.3 Wi-Fi module power consumption

Operating mode	Current consumption	
	Measured ($V_{cc} = 3.3\text{ V}$)	Expected
Tx mode	270 mA @ 18 dBm	212 mA @ 18 dBm
Rx mode	50 mA	40 mA
Sleep mode	1.5 mA	6 μA

pected value of 6 μA . To test battery autonomy, a 6 s transmission interval was used, for which the battery life was 20 h. All these tests were carried out with all the active sensors turned off.

74.6.8 Ambient Assisted Living Applications

Ambient assisted living (AAL) (Chap. 67) is an emergent technology that envisions offering new solutions for healthcare [74.31]. Those solutions will improve the quality of life of the elderly population and reduce costs associated with healthcare. However, before that happens, new solutions, including both hardware and software, must be available in order to acquire and store the required signals, to process and extract information from those signals, and to detect a set of features required to produce alarms and/or electronic assistance.

Despite the availability of several platforms for signal acquisition [74.49], a few technological issues must be solved before they can be used. The platforms must offer quality of service, be wearable, and operate for a comfortable period of time. Quality of service is re-

quired, otherwise no company will be able to offer such a highly critical service. When hiring a service for health monitoring, it must not fail, or the people involved must be aware of how safe the system is. Also, nobody will be willing to wear a device that reveals that he is being monitored, and it is also unpleasant to wear a device that needs to be recharged too often. At least, it must be able to operate for one full day. In this way, the system may be replaced when the person goes to bed or gets up.

Once the relevant data are available, an AAL system will allow remote monitoring of one, or a set of subjects, and their health and activity. Based on this and a set of information associated with each subject, it is possible to offer different levels of services. One will be self-reminders that are preset by the subject; the other will be reminders by some family member or caregiver; and finally, a health service based on clinical data may also be offered, if good enough health engines are available to extract the health condition.

Intelligence at Application Level

Figure 74.25 shows a snapshot of an application developed to collect data from several sensors, both biometric data (monitoring health and activity) and data from home automation [74.43]. Based on home automation, it is possible to infer the daily activity of a subject and help him in their routines.

This interface includes three main blocks. One block acts as a pill dispenser with the possibility to send text messages through the short message ser-

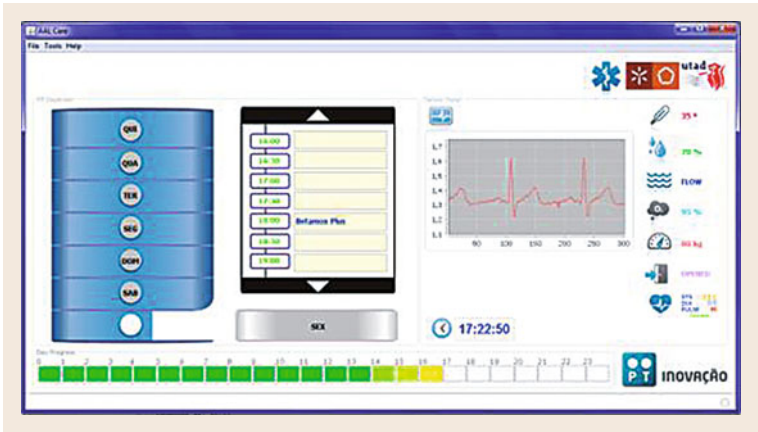


Fig. 74.25 Patient monitoring software snapshot

vice (SMS) and/or multimedia message service (MMS) with previously recorded familiar voices to provide advice regarding prescriptions. The second block provides a day (timeline) progress bar with emergency status (colors from green to red) linked to the continuous health monitoring system. The third block is a sensor panel showing data from several sensors for ECG, temperature, humidity, water flux, weight, door opening/closing, and arterial pressure with voice messages corresponding to high or very low arterial pressure readings.

Acquisition Hardware

The software operates based on the data acquired from the various sensors. Figure 74.26 shows the main blocks of the acquisition hardware. It is presented as worn by a person, but it is equally valid for data acquisition from home automation.

Since a few of those sensors must be mobile and must be placed on the person, the acquisition system

must be designed to have special features. It must be wireless, offer quality of service, have low power consumption, be wearable, and operate for at least 1 day on batteries. This led to the use of a modified version of a ZigBee-based platform.

For data acquisition, a small and power-efficient module was designed. Moreover, since the set of signals that may be relevant for this system are highly heterogeneous, the module was designed so that it can be configured for different applications. The most demanding signal in this scenario is the electrocardiogram, corresponding to measurement of the electrical activity of the heart muscle, which may range from about 0.5 mV to a maximum of 10 mV. It has a spectral range of 0.05–150 Hz for diagnostic ECG and 0.5–40 Hz for monitoring ECG.

The requirements of the front-end of an ECG sensor circuit include the capability to sense low-amplitude signals in the range of 0.05–10 mV, high common mode rejection ratio (CMRR), high input impedance,

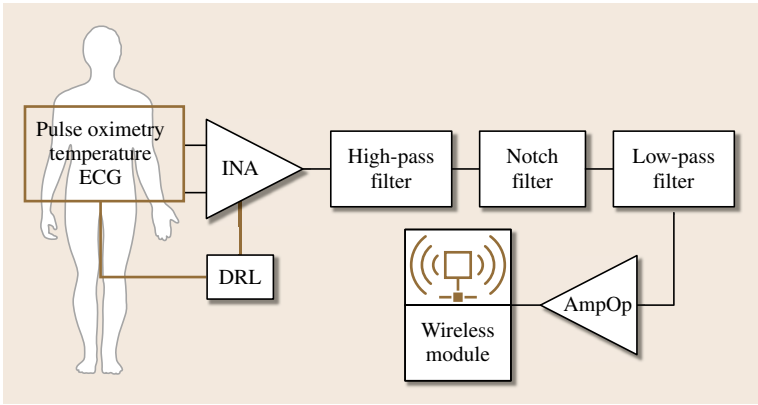


Fig. 74.26 Block diagram used for the data acquisition system (DRL – driven right leg; INA – instrumentation amplifier)

low input leakage current, flat response for 0.5–40 Hz, low weight, low power (<1 mA), and signal amplification over a range specified by the analog-to-digital converter (ADC).

The first stage of amplification is one of the most important components of the system. This amplifier presents excellent features for acquisition of biopotentials such as high input impedance, high differential gain, and high common mode rejection ratio. The integrated circuit (IC) used was the INA333 from Texas Instruments. The ECG is often contaminated with undesired spectral components. To attenuate these, the circuit must include a high-pass filter, notch filter, and a low-pass filter. All the implemented filters are active filters, so operational amplifiers were used. The ICs chosen to build the filters were the OPA2369 from Texas Instruments. The OPA2369 was chosen mainly because of its low quiescent current ($0.7\ \mu\text{A}/\text{channel}$), and low offset voltage ($250\ \mu\text{V}$). The high-pass filter must attenuate frequencies from zero to the cutoff frequency, thus it blocks the undesired DC components. The proposed filter is a second-order Sallen–Key high-pass filter with a Butterworth response. The filter's main features are its quality factor (Q) of 0.707, its unity gain, its cutoff frequency f_c of 0.45 Hz, and its slope of $-40\ \text{dB}/\text{decade}$. The ECG is frequently contaminated with 50/60 Hz power-line interference, or electrostatic charge. The strategy to attenuate this kind of interference involved a notch filter or band stop filter centered at 50 Hz. The implemented notch filter is a second-order Sallen–Key with gain of 1.5, Q of 1, f_c of 50 Hz, and slope of $-40\ \text{dB}/\text{decade}$. The low-pass filter is used to attenuate frequencies beyond the desired frequency for ECG measurement. These frequencies can come from sources such as radiofrequency and communication equipment. A second-order low-pass Sallen–Key filter with Butterworth response was implemented. The filter has unit gain, Q of 0.707, and cutoff frequency of 40 Hz. The filtered signals have to be amplified so that the ADC from the ZigBit module can measure it. The gain of this amplification stage is about 60, so the

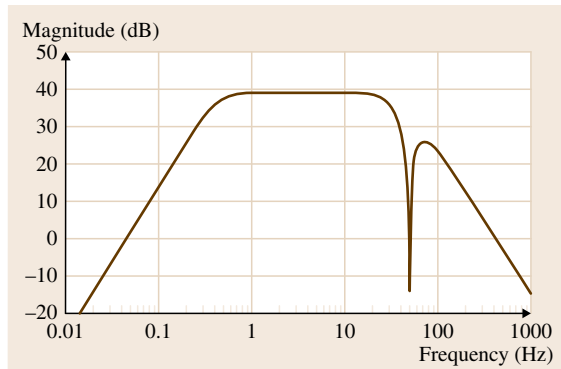


Fig. 74.27 Frequency response of the front-end electronic acquisition system

total gain of the system is 300. Finally, a driven right leg (DRL) circuit was also implemented. This circuit is used to amplify, invert, and feedback the common mode signal to the body, preventing saturation of the amplifiers, and reducing the output voltage due to CMRR. Figure 74.27 shows the frequency response for the described system.

Despite the low-power design, the system was designed to achieve high rejection of the 50 Hz component. This allows for a higher degree of system miniaturization, since when registering ECG signals with very close electrodes, interference may become a problem.

Since this is intended to be a flexible platform to acquire data and feed an AAL system, it was designed to be small and modular. Figure 74.28 shows the device dimensions.

The main achievement was the low current consumption for ECG acquisition ($90\ \mu\text{A}$) and the small size ($33 \times 30\ \text{mm}^2$). This module, combined with a transceiver module, allows acquisition of almost every signal produced by any transducers that may be found useful for use in AAL systems.

The developed platform was designed to acquire signals from a range of sensors, and to transmit them

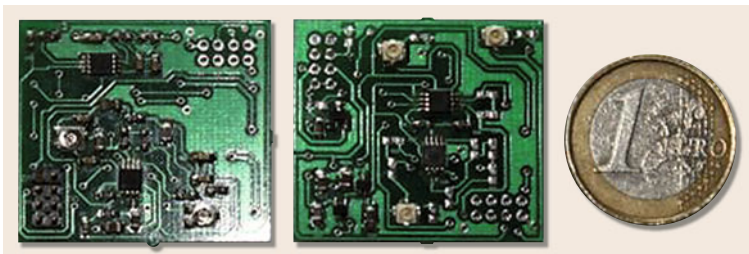


Fig. 74.28 Photograph of developed PCB layout

to a base station where the signals were registered and processed to monitor a subject during their daily life. The module is very low power, allowing operation for 1 day, and the wireless data were delivered with quality of service, making it suitable for AAL applications in controlled environments.

74.6.9 Wireless Link Design for Biomedical Applications

When a patient’s clinical state turns from a noncritical situation into a critical one, a context change occurs, and consequently, the healthcare network should adapt its performance requirements to this new situation [74.90]; for instance, higher monitoring activity and lower transmission delay of vital signals might be required when the clinical situation of a patient changes from non-critical to critical. Hence, healthcare networks should provide QoS facilities for e-emergency services, since these clearly demand high reliability, guaranteed bandwidth, and short delays.

A personal healthcare system consists of a group of sensors attached noninvasively to a patient for sensing of physiological parameters. These body sensor networks (BSN) have been used in hospitals during recent decades using conventional wired equipment, hence not allowing the patient to move around freely. However, recent advances in wireless sensor technology are changing this scenario by allowing mobile and permanent monitoring of patients, even during their normal daily activities. In such healthcare systems, the information sensed at the patient’s body is transmitted to a wireless base station, located no more than a few tens of meters away, and then delivered to a remote diagnosis center through a communication infrastructure.

Healthcare systems should be able to accomplish at least one crucial aim: to monitor a patient and, when an emergency occurs, immediately trigger an event to alert the patient and/or warn a remote caregiver. In this way, both the patient and the caregiver can ensure timely implementation of the appropriate procedure in accordance with the clinical episode. In addition, the system should be able to trigger an alert anticipating the case where the patient is unaware of the seriousness of his/her health situation.

Wireless sensor networks (WSN) are expected to be used in emergency healthcare applications in the near future. Since e-emergency systems should be totally reliable and efficient in order to provide valuable assistance to patients, quality of service (QoS) must be

present at distinct protocol levels, including the media access control (MAC) layer.

Vital Signals Requirements

Emergency medical care requires monitoring of several vital signals simultaneously; for example, a patient in an ambulance is monitored for blood pressure, heart and respiration rates, and temperature. Besides these primary signals, other information may be captured to help diagnosis and medical decision, such as electrocardiogram (ECG), blood glucose level, blood oxygen saturation, heart and breathing sounds, or even imaging in cases of trauma. All these data should be promptly accessible for comparison, computer-aided analysis, and decision-making. However, to achieve a more encompassing picture of the clinical situation, information about the patient’s personal characteristics (risk factors, degree of disease, age, etc.) and environmental context (e.g., in bed or mobile, alone or not, at work or at home, etc.) should also be provided [74.91].

Table 74.4 presents the electrical characteristics of the vital signals usually monitored in emergency medical care [74.92, 93] and Table 74.5 shows the range of typically monitored values.

If some signal exceeds its threshold, the local supervisor node should trigger an alarm to inform a caregiver or the patient himself. Table 74.5 presents typical

Table 74.4 Vital signal electrical characteristics

Vital signal	Frequency range (Hz)	Sampling rate (Hz)	Resolution (bits)
ECG (per lead)	0.01...60–125	120–250	16
Temperature	0...0.1–1	0.2–2	12
Oximetry	0...30	60	12
Arterial pressure	0...60	120	12
Respiration rate	0.1...10	20	12
Cardiac rate	0.4...5	10	12

Table 74.5 Alert detection parameters

Alert type	Detection parameter
Low SpO ₂	SpO ₂ < 90%
Bradycardia	HR < 40 bpm
Tachycardia	HR > 150 bpm
HR change	Δ HR / 5 min > 19%
HR stability	Max HR variability from past 4 reads > 10%
BP change	Systolic or diastolic change > ±11%

thresholds for SpO_2 , heart rate, and blood pressure signals for alert detection [74.94].

QoS Considerations for e-Health

Some authors argue that differentiation based on data priority is inherent to wireless sensor networks (WSN), since it is normal to have sensors to monitor distinct physical parameters simultaneously, just as happens in BSNs. Here, the importance of the collected information is necessarily distinct, and therefore the network must prioritize the transmission of critical data when a sudden clinical change in the patient occurs; for example, in patients with cardiac diseases, heart activity information is more important than body temperature data. Depending on the patient's clinical condition, the priority assigned to a vital signal can change dynamically.

Most current BSNs only offer best-effort service [74.95], which is not the most adequate for e-emergency. The given examples indicate that those networks require QoS support in critical cases. This would ensure adequate bandwidth for higher-priority streams for efficient data delivery, even in case of interference or fading.

QoS mechanisms are usually deployed in networks to guarantee consistent service levels concerning certain parameters, such as packet loss ratio, transmission delay, jitter, and available bandwidth. These are the traditional end-to-end parameters used to characterize the performance of any communication infrastructure, including BSNs; for instance, the total delay to an ECG signal being displayed in the monitor should be less than 3 s for useful real-time analysis by a cardiologist [74.95]. To guarantee that jitter does not affect the estimation of the R-wave fiducial point, which considerably modifies the spectrum, ECG signals require a minimum sampling rate of 250 Hz. It should be noted that no significant difference between ECG traces are detected when sampling the signal at rates between 250 and 500 Hz, but significant reduction in peak amplitude values and inaccurate interval measurements are obtained at 125 samples/s [74.96].

Approaches over than prioritization may additionally be used to provide QoS; for example, for efficiency reasons, a large packet length is chosen for noncritical situations. However, as soon as an emergency occurs, the packet size is reduced to meet the low-delay QoS requirement, and signals considered irrelevant to this emergency episode are sampled at a lesser rate, or not at all. Moreover, the computation power may be lowered to a minimum, since all data must be forwarded, as opposed to regular operation where, to save energy,

the cardiorespiratory rhythm can be computed onboard before sending it. Alternatively, an ECG signal could be processed in the sensor itself to extract its relevant features. In this way, only information about an event is transmitted (e.g., QRS features and the corresponding timestamp of the R-peak), hence reducing the traffic load and saving energy.

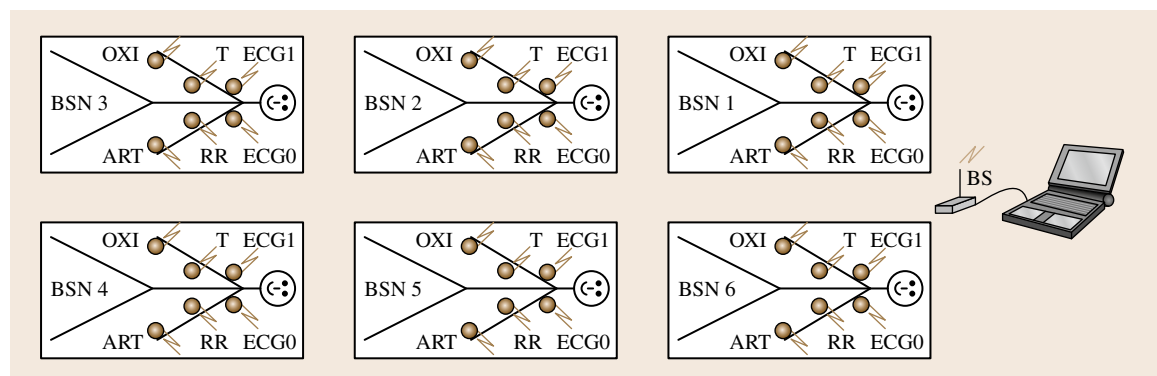
The vital signals captured from the patient's body must be delivered to a remote diagnosis and supporting center, through some available communication infrastructure, such as the Internet or a mobile cellular network. As a result, the delivered QoS depends necessarily on the network infrastructure chosen for the delivery of the mobile health services. Therefore, to meet the required QoS, the mobile health services platform needs to be able to acquire and use contextual information about the QoS offered by communication network infrastructures available at the patient's current location and time. High availability and reliability are the most desirable features that these network infrastructures should offer, as well as QoS guarantees for bandwidth, end-to-end delay, jitter, and loss.

e-Health Wireless Systems with QoS

Despite the number of healthcare systems already developed, few encompass QoS requirements. To assess how QoS support is being deployed in healthcare sensor networks, some projects were analyzed, as well as the QoS requirements that the respective authors considered important to incorporate in their implementations. Based on the related literature, QoS projects for e-health can be grouped according to the topics: (a) frameworks with QoS, (b) QoS through reconfiguration, and (c) algorithms for QoS enhancement [74.90].

The target deployment of each project is diverse, such as remote patient monitoring using standard protocols and mobile communication networks, mobile telemedicine under heavy traffic conditions, signal reconstruction, and BSN lifetime improvement. Thus, the QoS support approach is also distinct in each project: transmission priority scheduling along with data compression, self-reconfiguration of the BSN by data feedback, restoration algorithms to recover missing packets, and power management model for energy saving. Table 74.6 presents an overview of diverse QoS aspects that the authors have considered important to benefit their projects with, as well as the target environments where the projects are intended to be deployed. The target deployment of each project is diverse, such as remote patient monitoring using standard protocols and mobile communication networks,

Topic	Description	QoS goals	Target deployment
Frameworks with QoS	DiffServ based on transmission priority scheduling, data compression	Low delay and loss rate for vital signals, better bandwidth utilization	Remote medical applications using zigbee and mobile cellular networks
	Wireless diffserv infrastructure	Low loss rate, guaranteed bandwidth	Reliable network for mobile telemedicine under extreme traffic conditions
QoS through reconfiguration	Send feedback information into BSN to self-reconfiguration	Application level improvement	Remote and local patient monitoring
	Middleware system for reconfiguring bsn	Application level improvement	Increase BSN lifetime while respecting the qos needs
Algorithms for QoS enhancement	Restoration algorithms	Recover missing packets	Reconstruct a more functional signal for the doctor
	Power management model based on application states	Energy saving	Reserve power to the patient's more vital tasks



mobile telemedicine under heavy traffic conditions, signal reconstruction, and **BSN** lifetime improvement. Thus, the **QoS** support approach is also distinct in each project: transmission priority scheduling along with data compression, self-reconfiguration of the **BSN** by data feedback, restoration algorithms to recover missing packets, and power management model for energy saving.

Aware of the diverse QoS approaches presented above, Fig. 74.29 shows an experimental testbed to deploy QoS solutions based on a real clinical scenario [74.97]. The scenario is based on a hospital room containing six beds with one patient per bed. Each patient is monitored by a personal BSN, and one BS collects the vital signals of all patients. The vital signals being monitored are temperature (T, Temp), oximetry (OXI), arterial pressure (ART), respiration rate (RR), and ECG data. All monitored signals must be transmitted through a low-rate

According to the IEEE 1073 group, a wireless ECG electrode should generate 4 kbps of data, and the latency introduced by framing the data samples and the transmission delay should be below 500 ms. Since ECG signals are the most demanding in terms of QoS, we take this value as the maximum delay that any vital signal should have in this study. Continuous healthcare monitoring normally uses a three-electrode ECG device, therefore this situation is assumed in our study. According to Table 74.1, each BSN produces a maximum aggregated rate of 10 424 kbps, hence resulting in a maximum total traffic inside the hospital room of 62 544 kbps. Besides guaranteeing this minimum goodput and latency below 500 ms, the e-health system must also guarantee null packet loss for every vital signal, low energy consumption, and balanced energy drainage in every BSN.

ZigBee is a short-range, low-power, low-data-rate standard for wireless sensor networks that supports

a maximum data rate of 250 kbps in the 2.4 GHz band. Therefore, a ZigBee WSN is able to handle the whole traffic generated inside the hospital room without congestion. Nevertheless, other factors which may significantly affect QoS have to be considered, such as access to the wireless channel and errors which often occur during transmission.

Simulation results have shown that ZigBee is not adequate for several sensors to transmit ECG signals to a BS with full efficiency. The packet delivery efficiency starts to drop when three or more ECG devices belong to the same network channel. This is because ZigBee relies on carrier sense multiple access with collision avoidance (CSMA-CA), a contention-based MAC protocol that is vulnerable to collisions. Lost packets may be retransmitted, but there is a maximum number of five retries allowed by ZigBee before declaring channel access failure. ZigBee seems to be more adequate for BSNs that do not have large amounts of data to transfer with only several small data packets per hour, such as implanted medical sensors.

Diverse MAC protocols have been developed for wireless networks using contention-based or scheduled access techniques [74.98]. Traditional contention-based protocols assume that traffic is distributed stochastically. However, traffic in a WSN tends to be highly correlated and/or dominantly periodic, and so such protocols may not perform well in these conditions. Some CSMA-based protocols have been developed (e.g., Sensor-MAC (S-MAC) [74.98]) to help save energy in sensor network applications where nodes remain idle for long periods until detecting an event (e.g., surveillance), hence normally generating very little traffic in the network. Hence, it is inadequate to apply such MAC protocols in networks requiring high throughput and low latency, such as emergency biomedical applications. Time-division multiple access (TDMA) protocols divide time into slots that sensors can use to transmit data without needing to contend for the medium. Since quality of service is easier to assure in a collision-free environment than in a contentious-prone medium, and as a base station is available to keep the WSN scheduled and synchronized, a solution may be to use a TDMA-based MAC protocol to satisfy the QoS, such as the low-power, real-time protocol (LPRT) [74.99].

LPRT is a simple, beacon-based, low-power, real-time protocol that uses the available bandwidth dynamically and efficiently. Its highly grained superframe starts with the transmission of a beacon frame (B) by the BS, followed by the contention access period (CAP), also called the contention period (CP) [74.99]. The CAP

may be used for the (dis)association or configuration of a body sensor network (BSN). The contention-free period (CFP) follows the CAP. The CFP is composed of the normal transmission period (NTP) and the retransmission period (RP). The NTP is used for sensor nodes to transmit new data. Lost data are retransmitted in the RP. Data packets are sent in contiguous slots of the CFP. The attribution of slots in the CFP is announced through a list of allocation fields carried in the payload of each beacon. Each allocation field contains the association identification and the initial transmission slot (ITS) for every sensor node in the WSN. Data frames transmitted to the BS during the NTP are acknowledged by the ACK bitmap present in the beacon of the next superframe.

The beacon size may become relatively large, since it is directly dependent on the number of sensor nodes associated with the BS. A sensor node transmits data in the superframe only if the corresponding beacon carrying the list of allocation fields is received, and a single retransmission procedure is used in case of transmission failure. These characteristics may lead to a significant packet loss ratio if communications occur in a wireless channel with an appreciable bit error ratio (BER). To improve its robustness against bit errors, a solution based on short-size beacons has been proposed.

Proposals to Improve LPRT

To define a MAC protocol more robust to channel errors, solutions based on short-size beacons, autonomous transmissions, and multiple retransmissions have been added to LPRT. RP is defined after the CAP so that a packet is retransmitted away from an eventual burst error condition responsible for the transmission failure that occurred during the last NTP. RP is not placed after the NTP to avoid a sensor node transmitting in the NTP with a variable packet size.

Short-Size Beacons. To assure good performance of the e-emergency WSN, the percentage of lost beacons should be very low. A strategy to accomplish this goal is to send beacons with a convenient transmission power, since the bit error ratio (BER) of the channel decreases as the signal-to-noise ratio (SNR) increases. In addition, the beacon frame length should be as small as possible. So, whenever possible, the beacon payload contains only the ACK bitmap to acknowledge the frames correctly received during the NTP of the last superframe, and the CAP size of the current superframe (assuming that the start slot of the CAP is known). In this case, sensor nodes must run an algorithm to compute which slots

should be used to (re)transmit data without interfering with each other, in accordance with a predefined order schema. Using this strategy, the energy consumption in each BSN improves too, since smaller-size beacons are received by the sensor nodes.

According to the received ACK bitmap, each sensor node must calculate the corresponding superframe slots to transmit its data. If a sensor node does not receive a beacon or a short sequence of beacons, it may continue to send its new data in the NTP (i. e., autonomous transmissions), since a sensor node clock drift of the order of microseconds should permit the WSN to continue to be synchronized during a few consecutive beacon intervals. However, the sensor node cannot retransmit any data in the RP, because the ACK bitmap is not available and so it does not know how the RP slots are being allocated to the sensor nodes.

To save energy, it might be tempting to retransmit the lost data aggregated to the new data sent in the NTP, instead of retransmitting it in the RP. However such a strategy should be avoided, because the number of slots allocated in the NTP to each sensor node becomes variable. Consequently, if a sensor node does not receive the beacon, it has no way to know a priori which slots to use for transmission. Indeed, if a sensor node does not receive the ACK bitmap, it does not know which sensor nodes are going to transmit aggregated data in the current superframe, making it impossible to compute the new slot allocation schema. This situation does not occur if the NTP is used only to transmit new data packets. Hence, aggregated retransmissions in the NTP are not recommended, except in cases where aggregation does not imply taking more slots, such as for packets carrying temperature data.

Besides the ACK bitmap and the CAP size, a beacon may need to send reconfiguration instructions if a new clinical situation is detected in some BSN; for instance, higher monitoring activity and lower transmission delay of the vital signals might be required when a patient's clinical situation changes from noncritical to critical. In this case, the BS should inform all sensor nodes about the new situation and eventually reconfigure the WSN.

Retransmission in the RP. The retransmission order in the RP depends on the ACK bitmap received from the BS. Using an increasing slot sequence, firstly data of all sensor nodes having the highest priority and the bit false in the ACK bitmap are retransmitted successively. Then, data of all sensor nodes having the second highest priority and the bit false in the ACK bitmap are retransmitted successively, and so on. If there are

not enough slots to permit all required retransmissions, then schedule truncation is done to guarantee that no retransmission occurs in the NTP. It should be noted that retransmissions may not be the appropriate error recovery mechanism if losses are due to fading.

Retransmission in the CAP. If a sensor node does not receive the beacon, then it fails to receive the ACK bitmap, and therefore it does not know if the BS correctly received the packet sent in the NTP of the last superframe. Retransmitting that packet properly in the RP is impossible, because such a sensor node cannot compute the slot allocation schema of the RP. A solution to overcome this problem is to retransmit it during the CAP using the slotted CSMA-CA. This procedure should improve the packet loss ratio at the expense of some energy consumption and CAP slot wastage, since the sensor node may be transmitting a duplicated packet. Indeed, a packet already received by the BS may be retransmitted again in the CAP of the next superframe if the sensor node does not receive the beacon.

Simulation Testbed

To test the solutions proposed in the previous section and to compare them with LPRT, the Castalia simulator was used. For this purpose, a case study and distinct MAC protocol operation modes were programmed in Castalia. For each test run, a simulation time of 1 h was defined. Extending the simulation time would not significantly affect the results. During that time period, the BS sends around 16 363 beacons, carrying in the payload only the ACK bitmap (4B). It is assumed that the BER is equal in both communication directions.

In some graphics the probability P of a fully loaded packet being received at destination is used instead of the BER. The two parameters are related by

$$BER = 1 - P^{1/(8 \times MPS)}, \quad (74.1)$$

where MPS is the maximum physical frame size. Considering $MPS = 133$ B, the BER changes between 0 and around 6.5×10^{-4} as P decreases from 1 to 0.5 through the simulation runs. Typical values for BER in a good real wireless channel are of the order of 10^{-5} and in a bad channel may be less than 10^{-3} .

Operation Modes

Five distinct operation modes were defined for the simulation framework: 0, 1, 2, 3, and 4 [74.100]. In operation mode 0 there are no retransmissions in case of lost frames. This operation mode serves basically to define the limiting lines in the plots by considering the worst

case, corresponding to no error control mechanism in the system.

In operation mode 1 lost packets are retransmitted once at RP, and these retransmitted packets are not acknowledged. This operation mode implements basically the LPRT protocol.

In operation mode 2, packets with payload size above a certain threshold may be retransmitted during RP at most twice. The threshold was set at 40 bytes, so ECG and ART sensors have two chances to retransmit lost data. The second retransmission occurs only if the frame is not correctly received by the BS during the first retransmission. For this reason, the first retransmission must be acknowledged, although the second retransmission does not need to be. Packets with payload size below the defined threshold may be retransmitted once in the RP and are not acknowledged.

In operation mode 3, packets with payload size above a certain threshold may be retransmitted at most thrice in RP.

Finally, operation mode 4 is similar to operation mode 2, but if a sensor does not receive a beacon, it tries to send in the CAP the packet transmitted in the NTP of the last superframe. In all operation modes, frames transmitted in the NTP are acknowledged through the ACK table sent in the payload of the next beacon. No operation mode allows aggregated transmissions in the NTP or RP.

Compared with mode 1, the improvement achieved when the WSN operates in modes 2, 3, and 4 is clear from Fig. 74.30 and Fig. 74.31. These operation modes present ECG packet loss below 1% while the probability P is a little above 0.8. Such improvement is due to the multiple retransmission process occurring in the superframe. When the WSN operates in mode 3 instead of mode 4, the improvement in ECG traffic is significant for $P < 0.75$. However, for ART traffic it is almost irrelevant whether mode 3 or 4 is used. This is because the probability of losing an ART packet is lower than the probability of losing an ECG packet, as the packet size is smaller in the former.

Since the oximetry, respiration rate, and temperature sensors may only retransmit once in RP, no significant difference is noted in such traffic when operating in modes 1, 2 or 3 (Fig. 74.32). However, an improvement is noticed in oximetry and respiration rate traffic when operating in mode 4. Temperature traffic does not exhibit such improvement because it cannot make use of the CAP.

To better evaluate the cost of the various operating modes in terms of consumed energy, we firstly con-

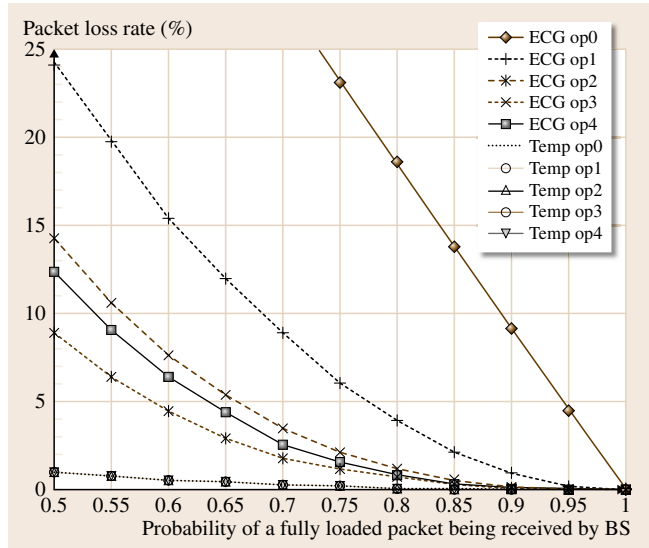


Fig. 74.30 Loss rate for ECG and temperature traffic

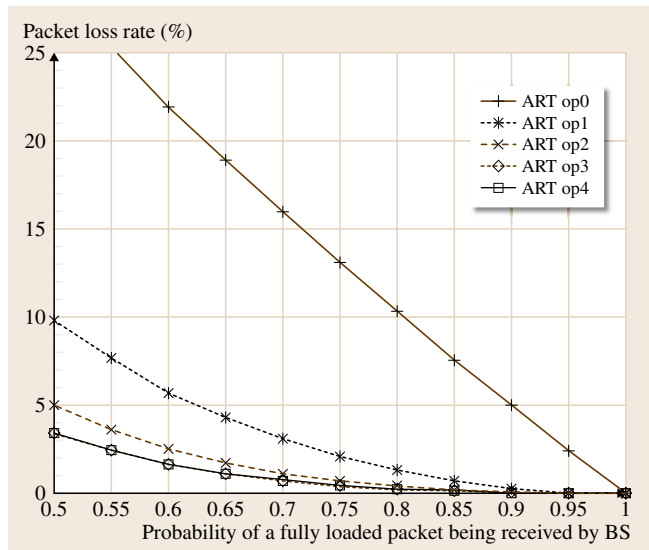


Fig. 74.31 Loss rate for ART traffic

sider that samplings performed by the sensing devices consume no power. Figure 74.33 presents for each operation mode the average lifetime per initial energy of the battery for each sensor type, relative to the simulations run with probability $P = 0.75$.

To determine how many minutes a type of sensor would live under such conditions, the values on the y-axis must be multiplied by the energy available in the battery when the sensor starts working. The available

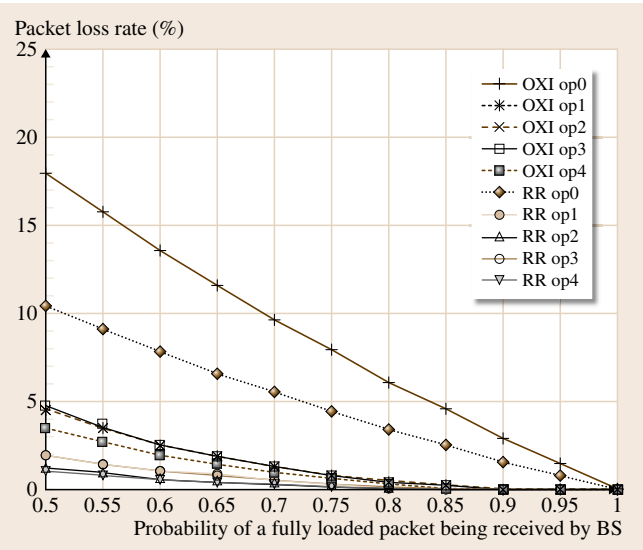


Fig. 74.32 Loss rate for OXI, and RR traffic

energy E in a battery with n cells, having U (volts) and capacity of C (mAh) for each cell, is $E = n U C \times 3.6 \text{ J}$; for instance, the 3 V, 40 mAh rechargeable Li polymer battery used by an ECG sensor node contains 432 J of initial energy before starting to power a sensor. So, according to Fig. 74.33, an ECG sensor operated in mode 4 and powered by that battery would live for $3.68 \times 432 \text{ min} \approx 26.5 \text{ h}$. For every sensor type, the energy consumption is not significantly increased if the probability $P > 0.75$ and the operation mode is 1, 2, 3 or 4.

Next, it was considered that each sampling performed by the sensing devices consumes 0.01 mJ of energy. Figure 74.34 presents the results for this situation. The effect of sampling rate on the lifetimes of the sensors is clear.

As ECG sensors have both the highest sampling rate and the largest transmitted packet size, the energy drainage is faster than for the other sensor types. The differences observed in each sensor type for the various operating modes are almost indistinguishable.

Conclusions

Since wireless e-emergency networks should be reliable and efficient, they need to support QoS at distinct protocol levels, including the MAC layer. The deployment of LPRT in an e-health system leads to low power consumption, controlled latency, and throughput efficiency. However, as simulations have proved, this approach lacks reliability if the wireless channel

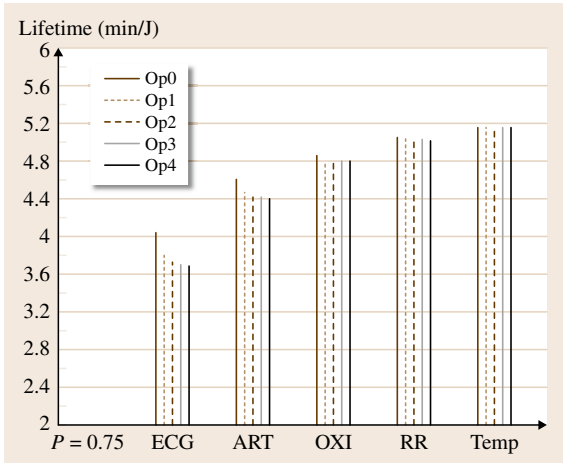


Fig. 74.33 Lifetime for sensing consumption of 0 mJ

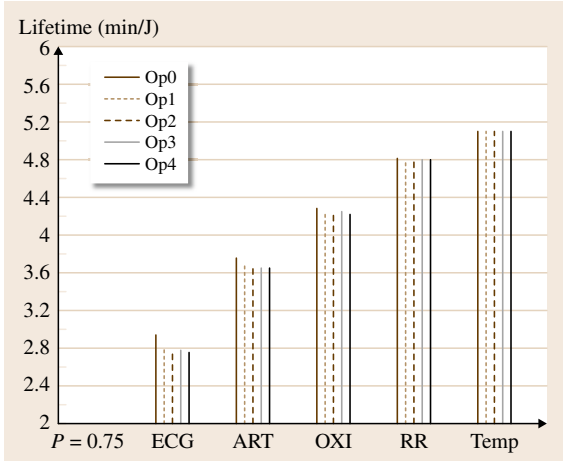


Fig. 74.34 Lifetime for sensing consumption of 0.01 mJ

presents a given bit error rate. To define a MAC protocol that is more robust to bit error conditions than LPRT, some solutions based on short-size beacons and multiple retransmissions were tested. The results showed that such strategies lead to meaningful improvements in packet loss rate, without significantly increasing energy consumption.

Despite presenting the best performance regarding the loss rate parameter, we foresee that a network operating in mode 3 may face scalability problems, since considerable bandwidth wastage occurs in the RP. Since operation mode 4 offers better scalability and a good compromise in terms of packet loss rate, we believe that an e-health network operating in this mode and using short-size beacons may present interesting performance.

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